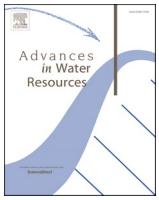
Advances in Water Resources 212 (2026) 105301



Contents lists available at ScienceDirect

Advances in Water Resources

journal homepage: www.elsevier.com/locate/advwatres





A novel analytical solution for quantifying seawater intrusion in coastal aquifers embedded with a cutoff wall

Jiazhi Sun [a,b], Wenlong Shi [c], Huiqiang Wu [a,b], Jie Yang [a,b], Chunhui Lu [a,b,*]

[a] State Key Laboratory of Water Disaster Prevention, Hohai University, Nanjing, China
[b] Yangtze Institute for Conservation and Development, Hohai University, Nanjing, China
[c] General Institute of Water Resources and Hydropower Planning and Design, Ministry of Water Resources, Beijing, 100120, PR China

ARTICLE INFO

Keywords:
Seawater intrusion
Coastal aquifer
Cutoff wall
Analytical solution

ABSTRACT

Cutoff walls are underground structures built to prevent seawater intrusion in coastal aquifers. Numerical and experimental models have been developed to study the effectiveness of cutoff walls in repelling seawater. Analytical solutions available for quantifying seawater intrusion in coastal aquifers embedded with cutoff walls require physical experiments to determine shape factors. This study presents a novel analytical solution capable of modeling seawater intrusion in coastal aquifers embedded with cutoff walls without the need of lab-experiments. The corrected analytical solution agrees well with the numerical and experimental solutions, confirming its accuracy. The results demonstrate that the construction of a cutoff wall can significantly raise the hydraulic head upstream of the wall. The effectiveness of the cutoff wall on controlling seawater intrusion increases with decreasing distance between the wall and the coastline, as well as decreasing opening height of the wall. When the cutoff wall is located further from the coastline, the effective range of the wall lies between two critical points derived from the analytical solutions. The analytical solutions developed in this study can serve as an independent tool for rapidly assessing the effectiveness of cutoff walls in preventing seawater intrusion.

1. Introduction

Fresh groundwater resources are frequently used for drinking, agricultural irrigation, and industrial usages in coastal regions worldwide. In many coastal areas, groundwater serves as the primary source of freshwater due to limited surface water availability and the high costs associated with alternative sources like desalination (Post et al., 2013). However, seawater intrusion exacerbated by rising sea levels and excessive groundwater extraction threatens these valuable freshwater resources (Ketabchi et al., 2016; Werner et al., 2013). Recent studies have revealed that sea-level rise accelerates the inland movement of seawater, while over-pumping of groundwater lowers the freshwater table, further promoting seawater intrusion into coastal aquifers (Foster et al., 2004; Liu et al., 2008; Michael et al., 2017). Additionally, land subsidence resulting from groundwater depletion can enhance the susceptibility of coastal regions to seawater intrusion (Herrera-García et al., 2021). The degradation of freshwater resources due to seawater intrusion poses significant challenges for sustainable water management in coastal regions, necessitating the development of effective mitigation and adaptation strategies.

In recent decades, various engineering strategies, such as hydraulic, air, and physical barriers, have been utilized to mitigate seawater intrusion. Hydraulic barriers, classified as positive, negative, or mixed, redistribute the hydraulic head using injection or pumping wells and have been successful in preventing seawater intrusion (Ebeling et al., 2019; Kacimov et al., 2009; Land et al., 2004; Lu et al., 2013, 2017; Luyun et al., 2011; Mahesha, 1996, 2001; Sheahan, 1977; Sherif and Hamza, 2001; Shi et al., 2020). However, obtaining freshwater resources for injection is challenging, particularly in regions with limited freshwater availability. Alternatively, air injection has been considered a viable option due to its availability and low cost (Sun and Semprich, 2013; Yang et al., 2024). Air barriers are created by injecting air into aquifer pores above the injection point, which reduces hydraulic conductivity and forms an air barrier. However, the implementation of air barriers in the field is still limited, particularly in confined aquifers where air entrapment is more likely (Dror et al., 2004).

Physical barriers, including fully penetrating barriers, subsurface dams, and cutoff walls, are widely used in many countries and have been extensively studied for their effectiveness in preventing seawater intrusion and enhancing groundwater extraction (Ishida et al., 2003;

* Corresponding author.
E-mail address: clu@hhu.edu.cn (C. Lu).

https://doi.org/10.1016/j.advwatres.2026.105301
Received 30 August 2025; Received in revised form 22 February 2026; Accepted 7 April 2026
Available online 7 April 2026
0309-1708/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Jamali et al., 2013; Japan Green Resources Agency, 2004; Kaleris and Ziogas, 2013; Luyun et al., 2009). Wu et al. (2020) indicated that fully penetrating barriers outperform cutoff walls and subsurface dams in terms of saltwater volume, seawater wedge toe position, and maximum safe pumping rate. However, fully penetrating barriers could entail substantial investments, trap residual saltwater, and lead to a significant rise in groundwater level (Dror et al., 2004; Zheng et al., 2020). Compared to fully penetrating barriers and subsurface dams, cutoff walls require less excavation depth and construction investments, making them a more suitable option for preventing seawater intrusion and minimizing the intense groundwater level rise.

Previous studies on the effectiveness of the cutoff wall in preventing seawater intrusion have primarily relied on numerical simulations and laboratory experiments (Abdoulhalik and Ahmed, 2017; Kaleris and Ziogas, 2013; Luyun et al., 2011; Wu et al., 2020). For example, Luyun et al. (2011) conducted 2D sandbox experiments and numerical simulations using the SEAWAT code to examine the effects of the cutoff wall location, penetration depth and recharge wells on seawater intrusion in unconfined aquifers. They found that a cutoff wall with deeper penetration and closer proximity to the coast exhibits a higher seawater repulsion ratio. Kaleris and Ziogas (2013) developed 3D numerical models based on SUTRA to investigate the effects of the cutoff wall on seawater intrusion and groundwater extractions. They demonstrated that the effectiveness of the cutoff wall depends on the wall depth, location, velocity ratio, mixing intensity, conductivity, and the relative conductivity of the wall. Abdoulhalik and Ahmed (2017) used numerical simulations and laboratory experiments to assess the effectiveness of the cutoff wall in multi-layered coastal aquifers. They found that a low-permeability layer at the bottom of the aquifer reduces the repulsion effect of the cutoff wall.

The study using analytical solutions to investigate the effect of cut-off walls on seawater intrusion is rare. Anwar (1983) derived an analytical solution to determine the shape of the freshwater-seawater interface in an unconfined aquifer with a cutoff wall. The simulated results were consistent with the vertical plane Hele-Shaw model. However, this analytical solution did not consider the mixing between freshwater and seawater. In addition, the solution included two shape factors that need to be determined experimentally (Kaleris and Ziogas, 2013), which implies that interface profiles can not be delineated without a Hele-Shaw model.

This study aims to develop a novel analytical solution for the freshwater-seawater interface subject to a cutoff wall embedded into an unconfined or confined aquifer, without the assistance of an experimental model. Numerical models based on the variable-density flow code SEAWAT are developed to validate the accuracy of the analytical solution. The effect of the cutoff wall on controlling seawater intrusion is assessed in terms of the length of the cutoff wall opening and its location. The findings are expected to establish a theoretical foundation for

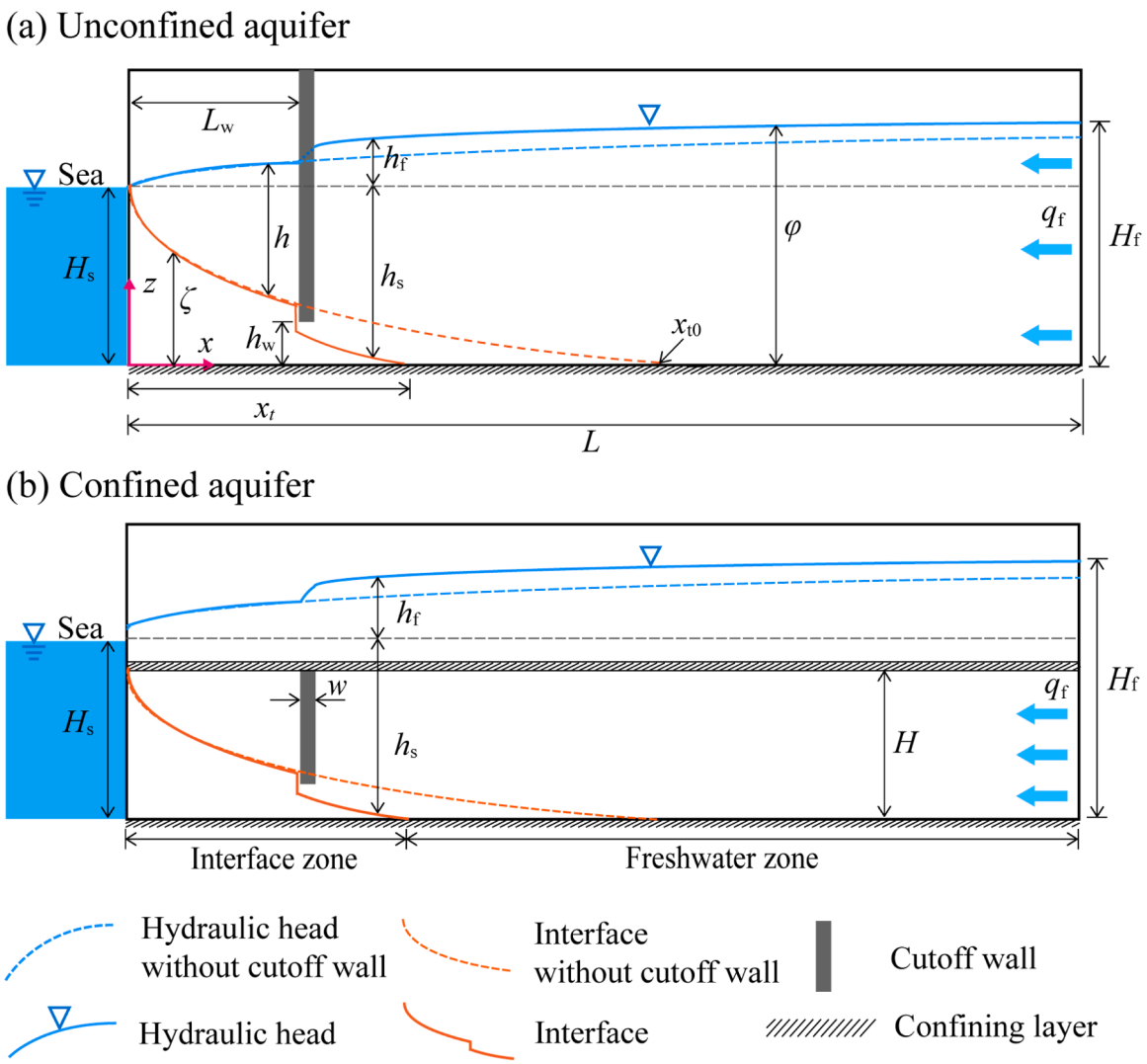


Fig. 1. Conceptual models of coastal aquifers with a cutoff wall: (a) unconfined aquifer, and (b) confined aquifer.

evaluating the performance of the cutoff wall in preventing seawater intrusion.

2. Methodology

2.1. Conceptual model

In this study, we consider both confined and unconfined aquifers with a cutoff wall. Fig. 1 illustrates the conceptual model of the idealized cross-sectional aquifers. The hydraulic conductivities of the aquifer and the cutoff wall are K_a [L/T] and K_w [L/T], respectively, with K_w being significantly smaller than K_a. The distance between the left edge of the cutoff wall and the coastline is denoted by L_w [L]. The width of the cutoff wall is denoted by w [L]. The height of the cutoff wall opening is represented by h_w [L]. The thickness of the confined aquifer is denoted as H [L]. The densities of freshwater and seawater are ρ_f [ML^{-3}] and ρ_s [ML^{-3}], respectively. The aquifer system with a cutoff wall is divided into the interface zone and the freshwater zone by the interface toe located at $x = x_t$, whereas x_{t0} denotes the interface toe location in the absence of the cutoff wall. The thickness of the freshwater flow path in the interface zone is h [L]. The height of the interface above the base is ζ [L]. The hydraulic head is denoted as φ [L] relative to the base of the aquifer. The sea level is represented by H_s [L]. The depth between the hydraulic head and the sea level is h_f [L], while the depth between the sea level and the interface is denoted as h_s [L]. A constant inland flux q_f [L^2T^{-1}] is assigned to the right-side boundary.

2.2. Mathematical model

The interface zone can be divided into sub-zones according to the cutoff wall location relative to the interface and the cutoff wall opening height. Six scenarios are defined according to the relative position of the interface toe and the cutoff wall, as summarized in Table 1 and illustrated in Fig. 2. Scenarios (a)–(c) illustrate situations where the cutoff wall does not intersect the interface (i.e., $h_w \geq \zeta(L_w)$, where $\zeta(L_w)$ denotes the interface elevation above the base at $x = L_w$). Scenarios (d)–(e) depict scenarios where the interface intersects the cutoff wall (i.e., $h_w < \zeta(L_w)$) and exits the wall at $x = x_h$, where x_h denotes the horizontal location at which the interface leaves the cutoff wall, while the interface leave the cutoff wall at $x = L_w + w$ in scenario (f).

To facilitate the derivation of the analytical solution, the interface flow zone in scenario (b) is divided into two sub-zones, corresponding to $x\langle L_w$ and $x\rangle L_w$, respectively. In scenario (c), the interface flow zone is divided into three sub-zones defined by $x\langle L_w$, $L_w < x < L_w + w$, and $x\rangle L_w + w$. In scenario (d), the interface flow zone is divided into three sub-zones defined by $x\langle L_w$, $L_w < x < x_h$, and $x\rangle x_h$. In scenario (f), the interface flow zone is divided into three sub-zones defined by $x\langle L_w$, $L_w < x < L_w + w$, and $x\rangle L_w + w$. In scenario (e), the interface flow zone is divided into four sub-zones defined by $x\langle L_w$, $L_w < x < x_h$, $x_h < x < L_w + w$, and $x\rangle L_w + w$.

Table 1
Scenario definitions based on the relative position of the interface toe and the cutoff wall.

Scenario	Condition	Description
(a)	$h_w \geq \zeta(L_w)$, $x_{t0} < L_w$	The cutoff wall is constructed further inland than the initial interface toe.
(b)	$h_w \geq \zeta(L_w)$, $L_w < x_t < L_w + w$	The interface toe is located beneath the cutoff wall.
(c)	$h_w \geq \zeta(L_w)$, $x_t > L_w + w$	The interface toe moves further inland than the cutoff wall.
(d)	$h_w < \zeta(L_w)$, $x_h < L_w + w$, $x_t < L_w + w$	The interface enters the cutoff wall and exits at x_h, with the interface toe located between L_w and $L_w + w$.
(e)	$h_w < \zeta(L_w)$, $x_h < L_w + w$, $x_t > L_w + w$	The interface enters the cutoff wall and exits at x_h, with the interface toe is located further inland than the cutoff wall.
(f)	$h_w < \zeta(L_w)$, $h_w < \zeta(L_w + w)$	The interface enters the cutoff wall and exits at the right boundary of the wall.

This study is based on the following assumptions: (1) the coastal aquifer system is in a steady-state condition; (2) freshwater and seawater are immiscible and separated by a sharp interface; (3) the Dupuit–Forchheimer approximation is adopted; and (4) the interface position is described by the Ghyben–Herzberg relation, with the seawater phase assumed to be static (Badon Ghyben, 1888; Herzberg, 1901). It should be noted that the Dupuit–Forchheimer approximation, which assumes predominantly horizontal freshwater flow, may be locally violated in the immediate vicinity of the cutoff wall. In this narrow region, abrupt constriction and expansion of the freshwater flow path can induce strong vertical hydraulic gradients and two-dimensional flow components. Consequently, detailed two-dimensional flow processes and vertical hydraulic gradients in the immediate vicinity of the wall are not explicitly resolved in the analytical formulation. Nevertheless, the approximation remains reasonable on the seaward and landward of the wall at the regional scale, where the freshwater thickness varies gradually. Under these assumptions, the one-dimensional governing equation for steady-state freshwater flow in a vertically layered coastal aquifer is written as (Strack, 1976; Strack and Ausk, 2015):

$$\frac{d}{dx}\left(T_i\frac{d\varphi_i}{dx}\right) = 0, \quad T_i = \sum K_j h_{i,j} \tag{1}$$

where the subscript i [-] is the sub-zone number, and j [-] = a or w is the layer index. φ_i [L] is the hydraulic head of the ith sub-zone. K_j [LT^{-1}] is the hydraulic conductivity in the ith sub-zone, of which the value is K_a [LT^{-1}] or K_w [LT^{-1}]. $h_{i,j}$ [L] is the thickness of the freshwater flow path in layer j within the ith sub-zone.

Based on the Ghyben-Herzberg formula, the relationship between the thickness of the flow path (i.e., h_i) and the hydraulic head (i.e., φ_i) is expressed as (Stack, 1976):

$$\begin{aligned} &\text{Confinedaquifers}: h_i = \alpha\varphi_i - (\alpha+1)H_s + H \\ &\text{Unconfinedaquifers}: h_i = (\alpha+1)(\varphi_i - H_s) \end{aligned} \tag{2}$$

where α [-] is the ratio of the freshwater density to the density difference between seawater and freshwater, defined as $\alpha = \rho_f / (\rho_s - \rho_f)$, with ρ_s and ρ_f denoting the densities of seawater and freshwater, respectively.

Neglecting the seepage face (i.e., $h = 0$ at the coast), the BC along the coastline ($x = 0$) is obtained by Eq. (2):

$$\begin{aligned} &\text{Confinedaquifers}: \varphi_s = \left(1 + \frac{1}{\alpha}\right)H_s - \frac{1}{\alpha}H \\ &\text{Unconfinedaquifers}: \varphi_s = H_s \end{aligned} \tag{3}$$

where φ_s denotes the hydraulic head at the coastline.

The inland constant flux BC ($x = L$) is expressed as:

$$K_a h_i \left.\frac{\partial\varphi_i}{\partial x}\right|_{x=L} = q_f \tag{4}$$

where q_f (positive) [L^2T^{-1}] is the inland flux per unit length of the coastline. As the aquifer system is in steady state with no source or sink terms, the freshwater flux through the aquifer is constant everywhere. The solution for a constant head inland BC (i.e., $\varphi = H_f$) is available via the relationship between the constant inland flux and hydraulic head based on the solution for the constant flux BC (Lu et al., 2015 and 2016).

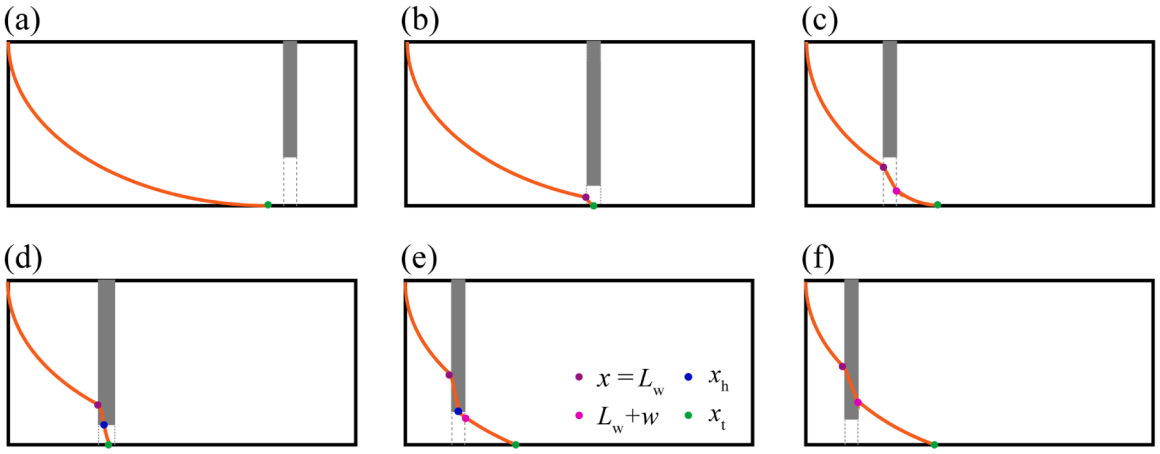


Fig. 2. Conceptual models of coastal aquifers categorized according to the cutoff wall location and length: (a) $h_w > \zeta(L_w)$ and $x_{t0} < L_w$, (b) $h_w > \zeta(L_w)$ and $L_w < x_t < L_w + w$, (c) $h_w > \zeta(L_w)$, and $x_t > L_w + w$, (d) $h_w < \zeta(L_w)$, $x_h < L_w + w$, and $x_t < L_w + w$, (e) $h_w < \zeta(L_w)$, $x_h < L_w + w$, and $x_t > L_w + w$, and (f) $h_w < \zeta(L_w)$, and $h_w < \zeta(L_w + w)$. The right side of each conceptual model is the inland boundary condition (BC), and the left is the seaward BC.

The continuity requirements of the hydraulic head at each of the zone boundaries ($x = l$) of two subzones can be written as:

$$\varphi_i(l) = \varphi_{i+1}(l) \tag{5}$$

where the value of l can be L_w, $L_w + w$, and x_h in different scenarios.

Based on Eq. (2), with $h = H$ for the confined case and $h = \varphi$ for the unconfined case, the relationship between the hydraulic head and the elevation of sea level at the interface toe location is given by:

$$\varphi(x_t) = \varphi_t = \left(1 + \frac{1}{\alpha}\right) H_s \tag{6}$$

Using the relation $\zeta = H - h$ in confined aquifers, and $\zeta = \varphi - h$ in unconfined aquifers, the elevation of the freshwater-seawater interface is expressed as:

$$\zeta = (\alpha + 1) H_s - \alpha\varphi \tag{7}$$

2.3. Analytical solution

2.3.1. Confined aquifer with constant flux inland BC

We define the following dimensionless variables and coefficients:

$$\begin{aligned} &\overline{\varphi}_i = \frac{\varphi_i - \varphi_s}{H}, \quad \overline{h}_w = \frac{h_w}{H}, \quad \overline{H}_s = \frac{H_s}{H}, \quad \kappa = \frac{K_w}{K_a}, \\ &\overline{x} = \frac{x}{x_t^a}, \quad \overline{L}_w = \frac{L_w}{x_t^a}, \quad \overline{w} = \frac{w}{x_t^a}, \quad \overline{x}_t = \frac{x_t}{x_t^a} \end{aligned} \tag{8}$$

where the overbar denotes a dimensionless symbol, and the superscripts *a–f* correspond to scenarios (a)–(f), respectively. Based on Eq. (8), analytical dimensionless solutions for the hydraulic head, interface toe location, and the limiting conditions of freshwater flow in coastal confined aquifers are derived from Eqs. (1)–(7). The detailed derivations of the corresponding dimensional solutions are provided in Appendix A.

The dimensionless hydraulic heads are expressed as:

$$\overline{\varphi}^a(\overline{x}) = \frac{1}{\alpha}\sqrt{\overline{x}} \quad 0 \le \overline{x} \le \overline{x}_t^a \tag{9}$$

$$\overline{\varphi}^b(\overline{x}) = \begin{cases} \overline{\varphi}^a(\overline{x}) & \overline{x} \le \overline{L}_w \\ \dfrac{1}{\alpha}\left(\sqrt{\overline{x} + \overline{\beta}_1\left(\overline{\beta}_1 - 2\sqrt{\overline{L}_w}\right)} + \overline{\beta}_1\right) & \overline{L}_w < \overline{x} < \overline{x}_t^b \end{cases} \tag{10}$$

where

$$\overline{\beta}_1 = (1 - \overline{h}_w)(1 - \kappa) \tag{11}$$

$$\overline{\varphi}^c(\overline{x}) = \begin{cases} \overline{\varphi}^b(\overline{x}) & 0 \le \overline{x} \le \overline{L}_w + \overline{w} \\ \dfrac{1}{\alpha}\sqrt{\overline{x} + 2\overline{\beta}_1\overline{\lambda}_1 + 2\overline{\beta}_1\sqrt{\overline{\lambda}_1^2 + \overline{w}}} & \overline{L}_w + \overline{w} < \overline{x} \le \overline{x}_t^c \end{cases} \tag{12}$$

where

$$\overline{\lambda}_1 = \overline{\beta}_1 - \sqrt{\overline{L}_w} \tag{13}$$

$$\overline{\varphi}^d(\overline{x}) = \begin{cases} \overline{\varphi}^a(\overline{x}) & \overline{x} \le \overline{L}_w \\ \dfrac{1}{\alpha}\sqrt{\dfrac{\overline{x} - (1-\kappa)\overline{L}_w}{\kappa}} & \overline{L}_w < \overline{x} \le \overline{x}_h \\ \dfrac{1}{\alpha}\left(\sqrt{\overline{x} - \dfrac{\kappa\overline{\beta}_1^2}{1-\kappa} - (1-\kappa)\overline{L}_w} + \overline{\beta}_1\right) & \overline{x}_h < \overline{x} \le \overline{x}_t^d \end{cases} \tag{14}$$

$$\overline{\varphi}^e(\overline{x}) = \begin{cases} \overline{\varphi}^d(\overline{x}) & 0 < \overline{x} \le \overline{L}_w + \overline{w} \\ \dfrac{1}{\alpha}\sqrt{\overline{x} - (1-\kappa)\overline{L}_w + \overline{\beta}_1\left(\dfrac{1-2\kappa}{1-\kappa}\overline{\beta}_1 + 2\sqrt{\kappa\overline{L}_w + \overline{w} - \dfrac{\kappa\overline{\beta}_1^2}{1-\kappa}}\right)} & \overline{L}_w + \overline{w} < \overline{x} \le \overline{x}_t^e \end{cases} \tag{15}$$

$$\overline{\varphi}^{f}(\overline{x})=\begin{cases}\overline{\varphi}^{d}(\overline{x}) & 0<\overline{x}\le\overline{L}_w+\overline{w}\\ \dfrac{1}{\alpha}\sqrt{\overline{x}-\left(1-\dfrac{1}{\kappa}\right)\overline{w}} & \overline{L}_w+\overline{w}<\overline{x}\le\overline{x}_t^{f}\end{cases}\tag{16}$$

The solutions for interface toe locations are:

$$\overline{x}_t^{a}=1\tag{17}$$

$$\overline{x}_t^{b}=1-2(1-\kappa)(1-\overline{h}_w)\left(1-\sqrt{\overline{L}_w}\right)\tag{18}$$

$$\overline{x}_t^{c}=1-2\overline{\beta}_1\left(\overline{\lambda}_1+\sqrt{\overline{\lambda}_1^2+\overline{w}}\right)\tag{19}$$

$$\overline{x}_t^{d}=\left(\overline{h}_w^2+\overline{L}_w\right)(1-\kappa)+\kappa\tag{20}$$

$$\overline{x}_t^{e}=(3-2\kappa)\kappa-\left(2\overline{\beta}_1\sqrt{\kappa\overline{L}_w+\overline{w}-\overline{\beta}_1^2\kappa/(1-\kappa)}-\overline{h}_w(2-\overline{h}_w)(1-\kappa)(1-2\kappa)-(1-\kappa)\overline{L}_w\right)\tag{21}$$

$$\overline{x}_t^{f}=1+\overline{w}\left(1-\frac{1}{\kappa}\right)\tag{22}$$

The limiting conditions for the six scenarios (see Section 2.2), respectively, are:

$$\begin{aligned}&\text{Scenarioa}: && 1\le\overline{L}_w\\ &\text{Scenariob}: && 0<1-\sqrt{\overline{L}_w}\le\overline{h}_w<1+\frac{\overline{L}_w+\overline{w}-1}{2(\kappa-1)\left(\sqrt{\overline{L}_w}-1\right)}\\ &\text{Scenarioc}: && 0<\max\left[1-\sqrt{\overline{L}_w},1+\frac{\overline{L}_w+\overline{w}-1}{2(\kappa-1)\left(\sqrt{\overline{L}_w}-1\right)}\right]\le\overline{h}_w<1\\ &\text{Scenariod}: && 0<1-\sqrt{\overline{L}_w+\frac{\overline{w}}{\kappa}}\le\overline{h}_w<\min\left[1-\sqrt{\overline{L}_w},\sqrt{\frac{\kappa(\overline{L}_w-1)+\overline{w}}{1-\kappa}}\right]\\ &\text{Scenarioe}: && 0<\max\left[1-\sqrt{\overline{L}_w+\frac{\overline{w}}{\kappa}},\sqrt{\frac{\kappa(\overline{L}_w-1)+\overline{w}}{1-\kappa}}\right]\le\overline{h}_w<1-\sqrt{\overline{L}_w}\\ &\text{Scenariof}: && 0<\overline{h}_w<1-\sqrt{\overline{L}_w+\frac{\overline{w}}{\kappa}}\end{aligned}\tag{23}$$

2.3.2. Unconfined aquifer with constant flux inland BC

Dimensionless variables are defined for freshwater flow in coastal unconfined aquifers with a cutoff wall, written as:

$$\begin{aligned}&\overline{\varphi}_i=\frac{\varphi_i-\varphi_s}{H_s},\quad \overline{h}_w=\frac{h_w}{H_s},\quad \kappa=\frac{K_w}{K_a}\\ &\overline{x}=\frac{x}{x_t^{u,a}},\quad \overline{L}_w=\frac{L_w}{x_t^{u,a}},\quad \overline{w}=\frac{w}{x_t^{u,a}},\quad \overline{x}_t=\frac{x_t}{x_t^{u,a}},\end{aligned}\tag{24}$$

where the parameter H_s is used to replace H in Eq. (8) for the nondimensional forms of φ_i and h_w. Based on Eq. (24), the dimensionless solutions of the hydraulic head for the freshwater flow model in unconfined aquifers are expressed as:

$$\overline{\varphi}^{u,a}(\overline{x})=\frac{1}{\alpha}\sqrt{\overline{x}}\quad 0\le\overline{x}\le\overline{x}_t^{u,a}\tag{25}$$

$$\overline{\varphi}^{u,b}(\overline{x})=\begin{cases}\overline{\varphi}^{u,a}(\overline{x}) & \overline{x}\le\overline{L}_w\\ \dfrac{1}{\alpha}\left(\sqrt{\dfrac{(\alpha+1)(\overline{x}-\overline{L}_w)}{\kappa+\alpha}+\overline{\lambda}_2^2}+\overline{\beta}_2\right) & \overline{L}_w<\overline{x}<\overline{x}_t^{u,b}\end{cases}\tag{26}$$

where

$$\overline{\beta}_2=\eta(1-\overline{h}_w),\quad \eta=\frac{1-\kappa}{\kappa+\alpha},\quad \overline{\lambda}_2=\overline{\beta}_2-\sqrt{\overline{L}_w}\tag{27}$$

$$\overline{\varphi}^{u,c}(\overline{x})=\begin{cases}\overline{\varphi}^{u,b}(\overline{x}) & 0\le\overline{x}\le\overline{L}_w+\overline{w}\\ \dfrac{1}{\alpha}\sqrt{\overline{x}+\eta\overline{w}+2\overline{\beta}_2\left(\overline{\lambda}_2+\sqrt{\overline{\lambda}_2^2+\dfrac{(1+\alpha)\overline{w}}{\kappa+\alpha}}\right)} & \overline{L}_w+\overline{w}<\overline{x}\le\overline{x}_t^{u,c}\end{cases}\tag{28}$$

$$\overline{\varphi}^{u,d}(\overline{x})=\begin{cases}\overline{\varphi}^{u,a}(\overline{x}) & \overline{x}\le\overline{L}_w\\ \dfrac{1}{\alpha}\sqrt{\overline{x}-(1-\kappa)\overline{L}_w} & \overline{L}_w<\overline{x}\le\overline{x}_h\\ \sqrt{\dfrac{\alpha+1}{\alpha}\left(\dfrac{\overline{x}-(1-\kappa)\overline{L}_w}{\alpha(\kappa+\alpha)}-\dfrac{\kappa\overline{\beta}_2^2}{1-\kappa}\right)}-\overline{\beta}_2 & \overline{x}_h<\overline{x}\le\overline{x}_t^{u,d}\end{cases}\tag{29}$$

$$\overline{\varphi}^{u,e}(\overline{x})=\begin{cases}\overline{\varphi}^{u,d}(\overline{x}) & 0<\overline{x}\le\overline{L}_w+\overline{w}\\ \dfrac{1}{\alpha}\sqrt{\overline{x}-\eta(\alpha)\overline{L}_w-\overline{w}-\dfrac{\alpha(\kappa-\alpha+2\kappa\alpha)\overline{\beta}_2^2}{1-\kappa}-2\alpha\overline{\beta}_2\sqrt{(1+\alpha)\left(\dfrac{\kappa\overline{L}_w+\overline{w}}{\kappa+\alpha}-\dfrac{\kappa\alpha\overline{\beta}_2^2}{1-\kappa}\right)}} & \overline{L}_w+\overline{w}<\overline{x}\le\overline{x}_t^{u,e}\end{cases}\tag{30}$$

$$\overline{\varphi}^{u,f}(\overline{x})=\begin{cases}\overline{\varphi}^{u,d}(\overline{x}) & 0<\overline{x}\le\overline{L}_w+\overline{w}\\ \dfrac{1}{\alpha}\sqrt{\overline{x}-\left(1-\dfrac{1}{\kappa}\right)\overline{w}} & \overline{L}_w+\overline{w}<\overline{x}\le\overline{x}_t^{u,f}\end{cases}\tag{31}$$

The expressions for the interface toe locations are:

$$\overline{x}_t^{u,a}=1\tag{32}$$

$$\overline{x}_t^{u,b}=\frac{2(\overline{h}_w-1)\left((\kappa-1)\sqrt{\overline{L}_w}-\kappa\right)\alpha+(2\overline{h}_w-1)\alpha+\kappa\left(1-\sqrt{\overline{L}_w}\right)+\sqrt{\overline{L}_w}}{\alpha+1}\tag{33}$$

$$\overline{x}_t^{u,c}=1-\frac{(1-\kappa)\overline{w}}{\kappa+\alpha}-2\overline{\beta}_2\left(\overline{\lambda}_2+\sqrt{\overline{\lambda}_2^2+\frac{(1+\alpha)\overline{w}}{\kappa+\alpha}}\right)\tag{34}$$

$$\overline{x}_t^{u,d}=\left(\frac{\alpha}{\alpha+1}\overline{h}_w^2+\overline{L}_w\right)(1-\kappa)+\kappa\tag{35}$$

$$\overline{x}_t^{u,e}=(\alpha+1)^2+\eta(\alpha\overline{L}_w-\overline{w})+\alpha\left(\frac{(\kappa-(1-2\kappa)\alpha)\overline{\beta}_2^2}{1-\kappa}-2\overline{\beta}_2\sqrt{(1+\alpha)\left(\frac{\kappa\overline{L}_w+\overline{w}}{\kappa+\alpha}-\frac{\kappa\alpha\overline{\beta}_2^2}{1-\kappa}\right)}\right)\tag{36}$$

$$\overline{x}_t^{u,f}=1+\overline{w}\left(1-\frac{1}{\kappa}\right)\tag{37}$$

The limiting conditions for the six scenarios (see Section 2.2), respectively, are:

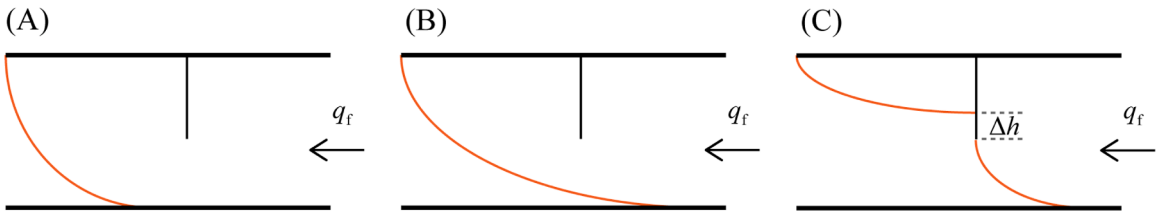


Fig. 3. The three options for the interface. In case C, a discontinuity in the interface occurs at the cutoff wall.

$$
\begin{aligned}
&\text{Scenarioa:} && 1 \le \overline{L}_w \\
&\text{Scenariob:} && 0 < 1-\sqrt{\overline{L}_w} \le \overline{h}_w < 1-\frac{1+\sqrt{\overline{L}_w}}{2\alpha\eta}+\frac{\overline{w}(1+\alpha)}{2(\kappa-1)(\sqrt{\overline{L}_w}-1)\alpha} \\
&\text{Scenarioc:} && 0 < \max\left[1-\sqrt{\overline{L}_w}, 1-\frac{1+\sqrt{\overline{L}_w}}{2\alpha\eta}+\frac{\overline{w}(1+\alpha)}{2(\kappa-1)(\sqrt{\overline{L}_w}-1)\alpha}\right] \le \overline{h}_w < 1 \\
&\text{Scenariod:} && 0 < 1-\sqrt{\overline{L}_w+\frac{\overline{w}}{\kappa}} \le \overline{h}_w < \min\left[1-\sqrt{\overline{L}_w}, \sqrt{\frac{\alpha+1}{\alpha}\frac{\kappa(\overline{L}_w-1)+\overline{w}}{1-\kappa}}\right] \\
&\text{Scenarioe:} && 0 < \max\left[1-\sqrt{\overline{L}_w+\frac{\overline{w}}{\kappa}}, \sqrt{\frac{\alpha+1}{\alpha}\frac{\kappa(\overline{L}_w-1)+\overline{w}}{1-\kappa}}\right] \le \overline{h}_w < 1-\sqrt{\overline{L}_w} \\
&\text{Scenariof:} && 0 < \overline{h}_w < 1-\sqrt{\overline{L}_w+\frac{\overline{w}}{\kappa}}
\end{aligned}
\tag{38}
$$

The detailed derivations of the dimensional solutions are shown in Appendix B.

2.3.3. Simplified analytical solution for impermeable cutoff walls with negligible thickness

As an alternative representation, a simplified analytical approximation is introduced for the limiting case of an effectively impermeable and thin cutoff wall. In this approximation, the cutoff wall is treated as effectively impermeable (i.e., $\kappa \approx 0$) and sufficiently thin such that its thickness can be neglected. To reflect these limiting conditions, a simplified analytical model for confined aquifers is derived by neglecting the flow details within the wall thickness and treating the wall as a sharp interface discontinuity. This simplification reduces the system to three primary cases, as shown in Fig. 3.

Case A corresponds to the defined scenario (a) in Section 2.3.1. The hydraulic head φ, and the interface toe location x_t are given by Eqs. (A3) and (A6), with their dimensionless forms provided in Eqs. (9) and (17), respectively.

Case B can be regarded as a special case of scenario (c) with $\kappa = 0$ and $\overline{w} = 0$. Substituting these limits into Eq. (23) yields $\sqrt{\overline{L}_w} > 1 - \overline{h}_w$, and substitution into Eq. (12) shows that the resulting expressions for Case B reduce to those of Case A. Therefore, in both cases A and B, the cutoff wall does not affect the interface.

When $\sqrt{\overline{L}_w} \le 1 - \overline{h}_w$, the system transitions to case C, which can be seen as the special case of scenario (e). In this case, the freshwater flux q_f is assumed to pass around the cutoff wall, while the detailed flow and interface in the vicinity of the wall are not explicitly considered. Instead, it is assumed that the Ghyben-Herzberg approximation still applies throughout the seawater phase. This means that a discontinuity in the hydraulic head $\Delta\varphi^C$ arises at the wall corresponding to the discontinuity in the height of the freshwater phase Δh^C given by:

$$
\Delta\varphi^C = \frac{1}{\alpha}\Delta h^C = \frac{1}{\alpha}\left((H-h_w) - \sqrt{\gamma_1 L_w}\right) \tag{39}
$$

where the superscript C denotes the case mark. In this case, h and φ on the seaward side of the wall are identical to those in cases A and B. On the landward side (i.e., $\overline{x} > \sqrt{\overline{L}_w}$) substitutions of $\kappa = 0$ and $\overline{w} = 0$ into Eqs. (15) and (21) give the dimensionless hydraulic head and interface toe location:

$$
\begin{aligned}
\overline{\varphi}^C &= \frac{1}{\alpha}\sqrt{\overline{x} - \overline{L}_w + (1-\overline{h}_w)^2} \\
\overline{x}_t^C &= 1 + \overline{L}_w - (1-\overline{h}_w)^2
\end{aligned}
\tag{40}
$$

For unconfined aquifers, cases A and B remain identical to scenario (a). For case C in unconfined aquifers, Eqs. (39) and (40) can be rewritten as:

$$
\Delta\varphi^{u,C} = \frac{1}{\alpha}\Delta h^{u,C} = \frac{1}{\alpha}\left((H_s - h_w) - \sqrt{\gamma_2 L_w}\right) \tag{41}
$$

Algorithm 1
Seawater intrusion with Cutoff Wall.

Step 1. Input Parameters
- Aquifer: H, H_s, K_a
- Cutoff wall: K_w, L_w, h_w, w
- Fluid: ρ_f, ρ_s
- Boundary: Inland Flux (q_f) OR Constant Head (h_f)

Step 2. Initialization and Pre-processing
1. Calculate Density Ratio: $\alpha = (\rho_s - \rho_f) / \rho_f$.
2. Boundary Condition Check:
   - **IF** input is Constant Head (h_f): Convert to flux q_f using Eq. (C6).
   - **ELSE**: Use input q_f.
3. Dimensionless Conversion:
   - **IF** Confined:
     - Calculate dimensionless variables using Eq. (8).
   - **IF** Unconfined:
     - Calculate dimensionless variables using Eq. (24).

Step 3: Scenario Identification *Evaluate the input-based inequalities to determine the physical scenario (a–f).*
- **IF** Confined:
  - Evaluate the limiting conditions listed in Eq. (23).
  - *Result:* Identify Scenario ($S \in \{a, b, c, d, e, f\}$).
- **IF** Unconfined:
  - Evaluate the limiting conditions listed in Eq. (38).
  - *Result:* Identify Scenario ($S \in \{a, b, c, d, e, f\}$).

Step 4: Analytical Calculation *Compute the dimensionless interface toe ($\overline{x}_t$) and hydraulic head ($\overline{\varphi}$) based on the identified Scenario S.*
- Toe Location ($\overline{x}_t$):
  - IF Confined: Use the specific equation from Eqs. (17–22).
  - IF Unconfined: Use the specific equation from Eqs. (32–37).
- Hydraulic Head ($\overline{\varphi}$):
  - IF Confined: Use the piecewise functions from Eqs. (9–16).
  - IF Unconfined: Use the piecewise functions from Eqs. (25)–(31).

Step 5: Dimensional Recovery and Output
1. Recover Dimensions: Convert $\overline{x}_t$ and $\overline{\varphi}$ back to dimensional x_t and φ using the definitions in Eq. (8) or Eq. (24).
2. Calculate Interface Elevation (ζ):
   - Compute interface profile using Eq. (7).
3. Apply Mixing Correction (Optional):
   - **IF** Flux-Controlled System: Adjust toe using Eq. (43).
   - **IF** Head-Controlled System: Adjust toe using Eq. (44).

Output:
- Interface Toe Location (x_t).
- Hydraulic Head Distribution ($\varphi(x)$).
- Interface Elevation Profile ($\zeta(x)$).

$$\overline{\varphi}^{u,C} = \frac{1}{\alpha}\sqrt{\overline{x} - \overline{L}_w + (1-\overline{h}_w)^2}$$
$$\overline{x}^{u,C} = 1 + \overline{L}_w - (1-\overline{h}_w)^2 \quad (42)$$

This simplified representation does not explicitly resolve detailed two-dimensional flow near the wall and should be interpreted as an analytical approximation.

2.4. Implementation procedure

To facilitate the practical application of the proposed analytical solution, the computational procedure is summarized in Algorithm 1. The algorithm outlines the key steps required to implement the analytical expressions and evaluate the hydraulic head distribution and interface toe location for a given cutoff wall configuration.

2.5. Numerical simulation and correction factor

The numerical variable-density model is developed based on SEAWAT-2000 (Langevin and Guo, 2006). The FloPy package is employed for pre- and post-processing the numerical models (Bakker et al., 2016). The numerical results depict the extent of the mixing zone between freshwater and seawater, using the 10% and 50% salt concentration contourlines to compare with the interface toe locations determined by the proposed analytical solution. The large-scale numerical simulations are performed for the confined aquifer. The modeled domain has the dimensions of 1000 m in length, 30 m in height, and 1 m in width. In all cases, a grid size of 1 × 1 × 1 m is used, resulting in 30,000 block elements. The hydraulic properties and solute transport parameters are detailed in Table 2. Specifically, K_a, D_m, and θ were taken from Pool and Carrera (2011); α_L and α_T from Lu et al. (2016); H, H_s, and q_f from Wu et al. (2020); and w from Chang et al. (2021). K_w is calculated based on the ratio between K_w and K_a reported by Kaleris and Ziogas (2013). L_w and h_w are specified in this study and systematically varied to quantify the effects of cutoff wall position and depth on seawater intrusion. The longitudinal dispersivity (α_L) is set at 1.0 m. The Peclet number, which is defined as the ratio of grid length and longitudinal dispersivity, equals 1.0. This value is lower than the standard value of 4.0, such that numerical dispersion can be neglected (Brovelli et al., 2007; Voss and Souza, 1987; Xie et al., 2010). The model is run for 20,000 days with a uniform time step of 1 day to ensure the steady state is reached.

Small-scale numerical simulations are also carried out for unconfined aquifers to compare the numerical results with both the proposed analytical solution and the experimental data reported by Luyun et al. (2011). The modeled domain ($L \times H \times W = 90 \times 42 \times 1$ cm) is discretized into fine grids with dimensions of 0.25 × 0.25 × 1 cm. The longitudinal dispersivity (α_L) is 0.1 cm. The Peclet number is 2.5, meeting the criterion of being below 4.0. This model setup is the same as the one used by Luyun et al. (2011). Other parameters are listed in Table 3.

Table 2
Hydraulic properties and solute transport parameters at the large-scale model.

Parameter	Value	Unit
Aquifer hydraulic conductivity (K_a)	10	m/d
Cutoff wall hydraulic conductivity (K_w)	0.001	m/d
Location of the cutoff wall (L_w)	0, 38, 77, 115	m
Height of the cutoff wall opening (h_w)	6, 12, 18, 24	m
Width of the cutoff wall (w)	1	m
Sea level (H_s)	35	m
Aquifer vertical thickness (H)	30	m
Freshwater discharge per unit width of coastline (q_f)	0.36	m^2/d
Freshwater density (ρ_f)	1000	kg/m^3
Seawater density (ρ_s)	1025	kg/m^3
Porosity (θ)	0.2	-
Molecular diffusion coefficient (D_m)	8.6×10^{-5}	m^2/d
Longitudinal dispersivity (α_L)	1.0	m
Transverse dispersivity (α_T)	0.1	m

Table 3
Hydraulic properties and solute transport parameters at the small-scale model.

Parameter	Value	Unit
Aquifer hydraulic conductivity (K_a)	1.34	cm/s
Cutoff wall hydraulic conductivity (K_w)	1.16×10^{-8}	cm/s
Location of the cutoff wall (L_w)	0, 11, 20, 35	cm
Height of the cutoff wall opening (h_w)	40, 29, 20, 5	cm
Width of the cutoff wall (w)	0.25	cm
Sea level (H_s)	40	cm
Aquifer vertical thickness (H)	42	cm
Inland constant head (h_f)	41.4	cm
Freshwater density (ρ_f)	1000	g/L
Seawater density (ρ_s)	1025	g/L
Porosity (θ)	0.4	-
Molecular diffusion coefficient (D_m)	1.16×10^{-6}	cm^2/s
Longitudinal dispersivity (α_L)	0.1	cm
Transverse dispersivity (α_T)	0.01	cm

As dispersivity can not be considered in the sharp interface model, Pool and Carrera (2011) introduced a correction factor to account for the mixing effect arising from dispersivity, expressed as:

$$\alpha' = \alpha\left[1 - \left(\frac{\alpha_T}{H}\right)^{1/6}\right]^{-1} \quad (43)$$

Eq. (43) was originally derived for seawater intrusion in flux-controlled systems. Lu and Werner (2013) proposed replacing the exponential of 1/6 with 1/4 to improve the accuracy of determining the interface toe location in head-controlled systems. Consequently, the correction factor is expressed as:

$$\alpha' = \alpha\left[1 - \left(\frac{\alpha_T}{H}\right)^{1/4}\right]^{-1} \quad (44)$$

Here H denotes the aquifer thickness of confined aquifers. For unconfined aquifers, where the saturated thickness varies spatially, the sea level H_s is adopted as an approximate thickness scale when applying the correction.

3. Results and discussion

3.1. Validation of analytical solution

This section demonstrates the validation of the proposed analytical solution incorporating the correction factor. Both confined and unconfined aquifers are considered. Fig. 4 shows the results of the corrected analytical solution (blue line) and the numerical solution (graduated color map and green lines). Since the inland BC is specified as a constant flux, Eq. (43) is employed for the correction. The corrected analytical solution using Eq. (43) generally agrees well with 50% concentration contour lines of the numerical solution. For completeness, validation results obtained under zero-dispersion conditions are provided in Appendix D, where the analytical solution is evaluated independently of dispersion effects. Nevertheless, discrepancies are observed between the analytical and numerical solutions (e.g., panels b1, c1, and d1) when the cutoff wall intercepts most of the freshwater flow path. As shown by the two-dimensional hydraulic head distributions (Fig. 5), obtained directly from SEAWAT simulations, these configurations are characterized by a large hydraulic head difference across the cutoff wall, generating strong vertical hydraulic gradients and enhanced vertical flow components in the cutoff region. Under such conditions, the Dupuit–Forchheimer approximation is locally violated, leading to larger deviations in the interface configuration. The performance of the cutoff wall in preventing seawater intrusion increases with the increasing cutoff wall height, as shown in panels (a1)–(a4), (b1)–(b4), (c1)–(c4), and (d1)–(d4). This is because the hydraulic head upstream of the cutoff wall rises, leading

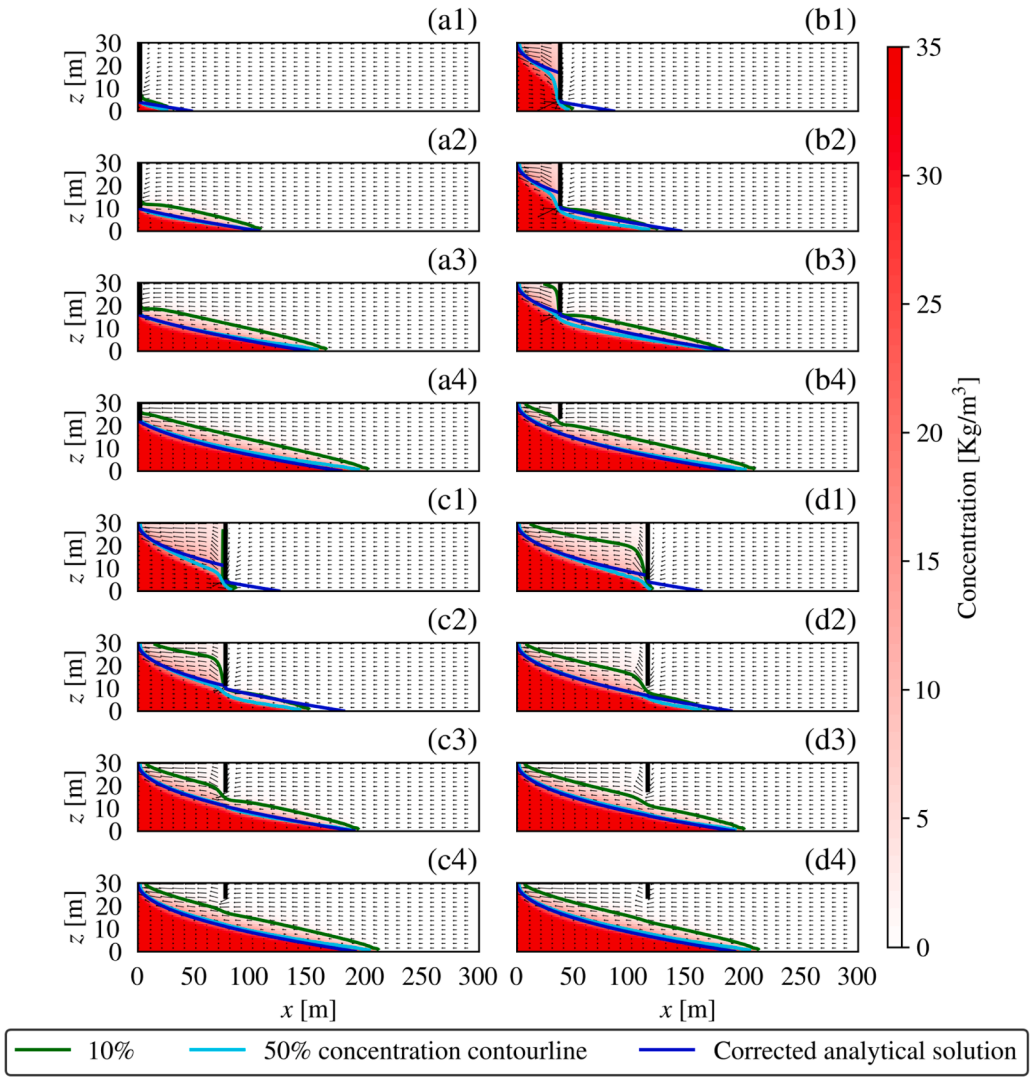


Fig. 4. Comparison between corrected analytical solutions (based on Eq. (43)) and numerical solutions in a confined aquifer after cutoff wall construction. Subplots are grouped by L_w, denoted by labels (a)–(d), and by h_w, denoted by indices 1–4. Labels (a)–(d) correspond to $L_w = 0$, 38 m, 77 m, and 115 m, respectively, while indices 1–4 represent $h_w = 6$, 12, 18, and 24 m.

to a steeper hydraulic gradient and an enhanced ability to inhibit seawater intrusion. Cutoff walls constructed closer to the coastline exhibit superior performance in preventing seawater intrusion, particularly those built directly on the coastline (i.e., panels a1-a4). For example, with a cutoff wall height of 18 m (i.e., panels a2, b2, c2, and d2), the interface toe locations are 106.0, 144.0, 181.0, and 188.3 m for the cases of $L_w = 0$, 38, 77, and 115 m, respectively, compared to $x_t^a = 191.7$ m in the absence of a cutoff wall. This is because the hydraulic head across the entire aquifer increases. Furthermore, the hydraulic head in the case of the cutoff wall at the coastline is the highest (see Fig. 4), leading to the most significant hydraulic gradient and the best performance in preventing seawater intrusion.

Fig. 6 presents the results of laboratory experiments, and analytical and numerical solutions at the laboratory scale. Experimental data were digitized from published plots in Luyun et al. (2011) using the WebPlotDigitizer tool (Rohatgi, 2021). During the experiments, cutoff walls were installed at a fixed location of $L_w = 20$ cm in an unconfined aquifer with a total height of 40 cm, while the cutoff wall heights were set to 11, 20, and 35 cm. This resulted in corresponding opening heights $h_w = 29$, 20, and 5 cm, as shown in Figs. 6b – 6d The initial condition is represented as $h_w = 40$ cm, as depicted in Fig. 6a. The experiments utilized an inland constant head BC. Therefore, we adjusted the current solution from an inland constant flux BC to a constant head BC, following the procedures outlined by Lu et al. (2016). Eq. (44) was then applied to correct for the head-controlled aquifer. Additional details are available in Appendix C. According to Luyun et al. (2011), the value of inland head in the analytical and numerical models is adjusted to 41.4 cm to enhance the agreement between these models and the experimental results. In general, the current analytical solution aligns closely with the experimental data concerning the interface toe location. A notable discrepancy of the interface profile exists at the cutoff wall location (i.e., $x = 20$ cm), where the experimental data exhibit irregularities in the otherwise smooth shape of the interface. Such irregularities were also reported in the laboratory experiments of Luyun et al. (2011), who attributed them to the absence of glass beads inside the slot used for installing the cutoff wall. The distinction between the analytical and numerical solutions arises from the neglect of factors such as the outflow face at the coast and vertical flow.

3.2. *Effects of the cutoff wall on interface toe location*

In this section, the effects of the cutoff wall on the interface toe location $\bar{x}_t$ are analyzed based on the derived analytical solution. Fig. 7 illustrates the dimensionless interface toe location. In panel (a), $\bar{x}_t$ is

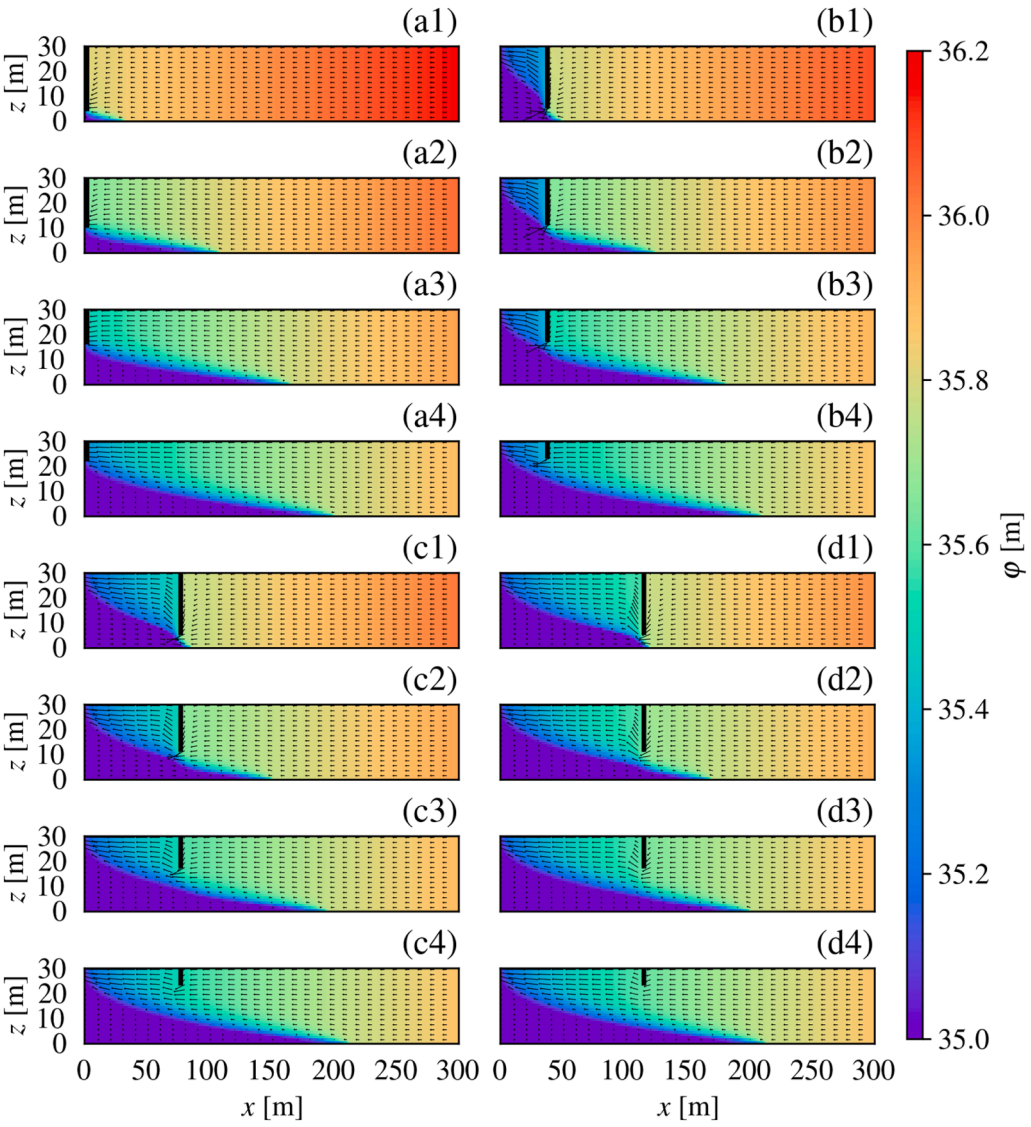


Fig. 5. Hydraulic head distributions obtained from numerical simulations for all investigated cases. The black lines indicate the location of the cutoff wall.

plotted as a function of the dimensionless cutoff wall opening height $\overline{h}_w$ for different dimensionless cutoff wall locations $\overline{L}_w = 0$, 0.2, 0.4, 0.6, and 0.8. In panel (b), $\overline{x}_t$ is shown as a function of $\overline{L}_w$ for different values of $\overline{h}_w$ = 0, 0.2, 0.4, 0.6, and 0.8, for both confined and unconfined aquifers. The nondimensional parameters $\overline{w} = 0.00052$ and $\kappa = 0.0001$ are held constant, based on the parameter values given in Table 2. Although the analytical expressions for confined and unconfined aquifers are not identical, they are very similar in form, and therefore produce overlapping dimensionless curves in panels (a) and (b) for the parameter values considered. Panel (a) shows that $\overline{x}_t$ remains nearly constant for small values of $\overline{h}_w$, followed by a substantial increase as $\overline{h}_w$ increases. When $\overline{h}_w$ is small, freshwater flow around the cutoff wall is dominated by the low-permeability cutoff wall, so $\overline{x}_t$ is only weakly sensitive to changes in $\overline{h}_w$. With increasing $\overline{h}_w$, an increasing proportion of the freshwater flux passes through the high-permeability aquifer, resulting in a noticeable inland shift of the interface toe. Eventually, the curves with varying $\overline{L}_w$ values converge into a single curve where $\overline{x}_t = 1$, indicating that the cutoff wall is incapable of preventing seawater intrusion. Such results indicate that scenario (d) is encountered when the values of $\overline{h}_w$ are relatively small, while scenario (c) emerges when $\overline{h}_w$ values are relatively large. For every curve with $\overline{L}_w > 0$, we define two critical points, and for the curves with $\overline{L}_w = 0$, we identify one critical point. These critical points are used to separate the $\overline{x}_t$-$\overline{h}_w$ curves into three or two segments. The left critical points representing the transition from scenario (d) to scenario (e) are expressed as $\overline{h}_w = \sqrt{\frac{\kappa(\overline{L}_w-1)+\overline{w}}{1-\kappa}}$ for confined aquifers and $\overline{h}_w = \sqrt{\frac{(\alpha+1)[\kappa(\overline{L}_w-1)+\overline{w}]}{\alpha(1-\kappa)}}$ for unconfined aquifers according to Eqs. (23) and (38). These critical points are indicated by grey lines. Although varying $\overline{L}_w$ leads to slightly different values of the left critical points, the resulting variation is minimal, with values ranging between 0.0714 and 0.0720. Consequently, the corresponding points appear nearly coincident at the scale of the figure. The right critical points are the transition points (indicated by dashed lines), beyond which $\overline{x}_t$ reaches a plateau (i.e., $\overline{x}_t = 1$) and remains constant regardless of further changes in $\overline{h}_w$. The critical values, $\overline{h}_w = 1 - \frac{\overline{L}_w+\overline{w}}{2\sqrt{\overline{L}_w(1-\kappa)}}$ and $\overline{h}_w = 1 - \frac{\alpha(\overline{L}_w+\overline{w}+\kappa)+\overline{w}}{2\sqrt{\overline{L}_w\alpha(1-\kappa)}}$, for confined and unconfined aquifers are obtained by differentiating Eqs. (19) and (34) with respect to $\overline{h}_w$ and setting the derivatives to zero, which corresponds to identifying points where the slope of the interface toe location with respect to $\overline{h}_w$ vanishes. This reflects that the cutoff wall can still prevent seawater intrusion even without directly intersecting the interface. The curves with $\overline{L}_w$=

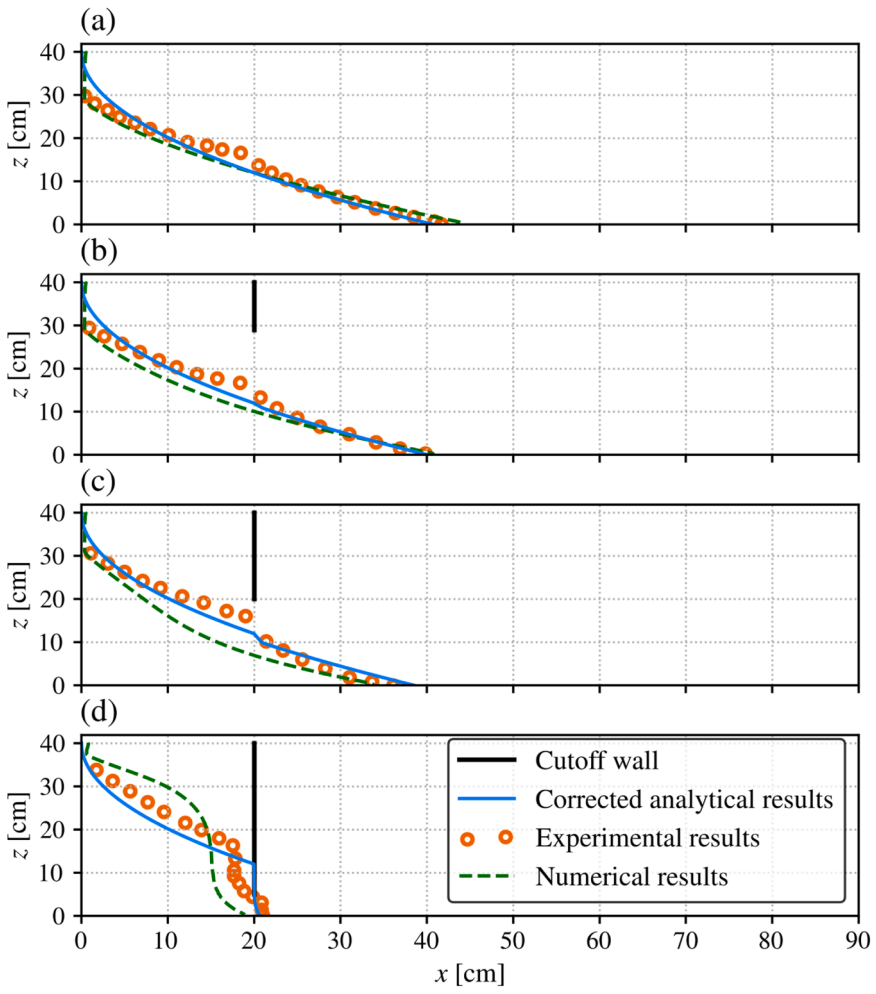


Fig. 6. Comparison among corrected analytical results (based on Eq. (44)), numerical simulations, and experimental data of Luyun et al. (2011) with the cutoff wall located at $L_w = 20$ cm. (a) $h_w = 40$ cm, (b) $h_w = 29$ cm, (c) $h_w = 20$ cm, and (d) $h_w = 5$ cm. The numerical results are represented by the 10%-salinity line (green and dashed line).

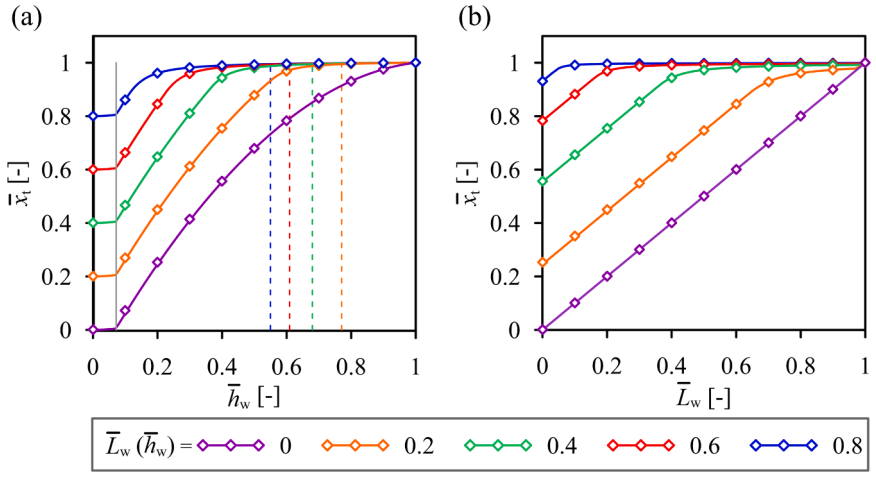


Fig. 7. Dependence of the dimensionless interface toe location $\bar{x}_t$ on cutoff wall parameters for confined and unconfined aquifers. (a) Dependence of $\bar{x}_t$ on the dimensionless cutoff wall opening height $\bar{h}_w$ for different dimensionless cutoff wall locations $\bar{L}_w = 0$, 0.2, 0.4, 0.6, and 0.8. (b) Dependence of $\bar{x}_t$ on the dimensionless cutoff wall location $\bar{L}_w$ for different values of $\bar{h}_w = 0$, 0.2, 0.4, 0.6, and 0.8. The dashed lines indicate the critical conditions at which cutoff walls at different locations fail to control seawater intrusion, with dashed lines of the same color corresponding to the same value of $\bar{L}_w$.The grey line marks the left critical point separating scenario (d) and scenario (e).

0 exclusively have the left critical points because they correspond to scenario (e) when $\overline{h}_w$ values are large, as described by Eqs. (21) and (36), which have no zero points. Panel (b) demonstrates that $\overline{x}_t$ and $\overline{L}_w$ are linearly related in the early stages, particularly when $\overline{h}_w = 0$. The linear relationship evolves into a nonlinear one as the values of $\overline{h}_w$ increase. Eventually, the curves converge into a single curve. The motivation behind these results is that the relationship between the cutoff wall and the interface shifts from scenario (e) to scenario (c) as $\overline{L}_w$ increases.

3.3. Effects of the cutoff wall on hydraulic head

Fig. 8 illustrates the effects of the dimensionless cutoff wall opening height $\overline{h}_w$ and the dimensionless cutoff wall location $\overline{L}_w$ on the dimensionless hydraulic head $\overline{\varphi}$ within the confined aquifer, with$\overline{w} = 0.00052$ and $\kappa = 0.0001$. These findings are also relevant to unconfined aquifers, as the nondimensional formulation reduces the differences between the confined and unconfined analytical solutions, leading to very similar results under the conditions considered. As shown in panel (a), $\overline{\varphi}$ increases upstream of the cutoff wall (i.e., $\overline{x}>\overline{L}_w + \overline{w}$) as $\overline{h}_w$ decreases for the given $\overline{L}_w = 0.3$. This can be attributed to the higher hydraulic gradient induced to compensate for the reduced $\overline{h}_w$ in order to maintain a constant freshwater discharge. The upper limit value of 0.8 is selected for $\overline{h}_w$ because the curves at 0.7 and 0.8 coincide, indicating that $\overline{\varphi}$ is independent of $\overline{h}_w$ when $\overline{h}_w > 0.7$. The precise critical point that distinguishes a functional cutoff wall from an ineffective one is approximately $\overline{h}_w = 0.72$ when $\overline{L}_w = 0.3$. The precise critical points of other cases are shown in Fig. 7. The effect of $\overline{L}_w$ on $\overline{\varphi}$ given $\overline{h}_w = 0.3$ is demonstrated in panel (b). Cutoff walls with smaller values of $\overline{L}_w$ result in a more pronounced increase in $\overline{\varphi}$. When $\overline{L}_w > 0.7$, the cutoff wall exerts only a weak influence on $\overline{\varphi}$. The curves in Fig. 8 show abrupt changes at the cutoff wall location due to the sudden reduction in flow path thickness. Because the flow path thickness naturally increases landward as a result of the interface geometry, the hydraulic head also increases from the coastline toward the inland boundary. When the cutoff wall is placed closer to the coastline, this reduction in flow path thickness occurs where the background flow path is narrower, resulting in a larger increase in hydraulic head that propagates landward.

3.4. Limitations and future opportunities

Based on the results presented above, the inland boundary conditions adopted in this study, including constant flux and constant head, represent idealized steady-state approximations of coastal aquifer systems. In reality, inland boundaries are often subject to temporal variability due to tidal oscillations, seasonal recharge, and groundwater abstraction. Such temporal variations may induce transient responses in hydraulic head and interface position that are not captured by the present steady-state formulation. The analytical solution developed here is therefore intended to describe the long-term or time-averaged behavior of seawater intrusion under given boundary conditions. It provides a first-order estimate of the interface toe position and the relative effectiveness of cutoff walls, while transient fluctuations may be superimposed on this baseline behavior.

Another important limitation of the present analytical framework is

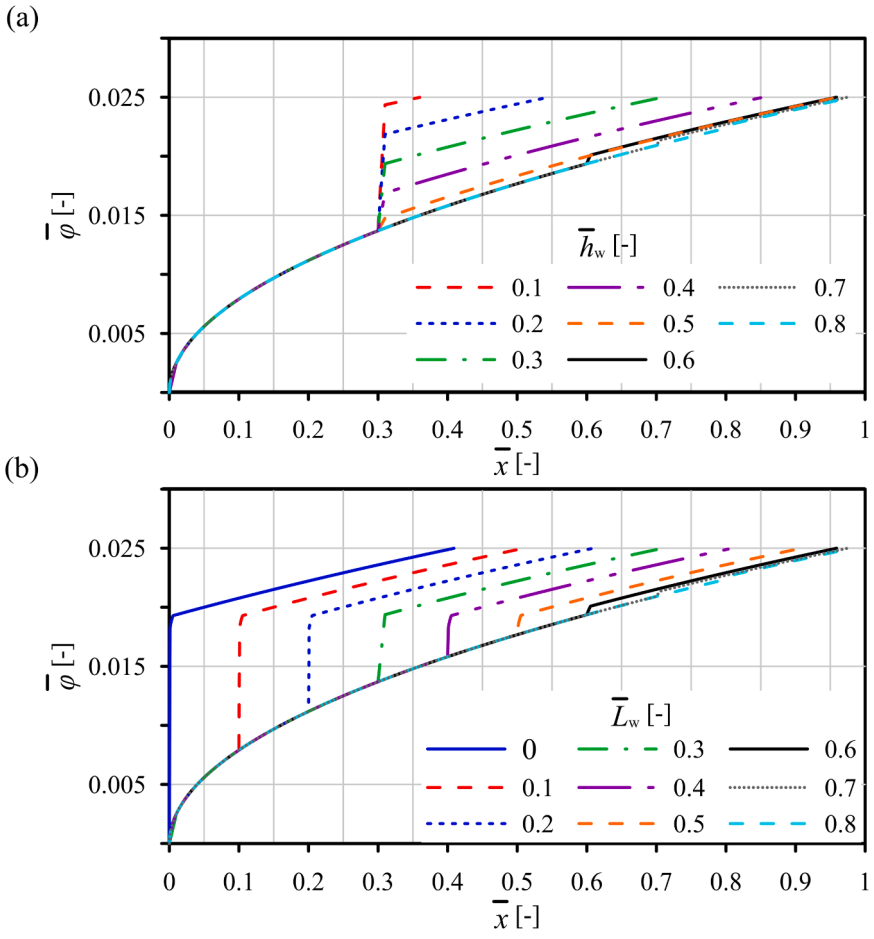


Fig. 8. Dimensionless hydraulic head distribution in a confined aquifer. Effects of (a) the dimensionless cutoff wall opening height $\overline{h}_w$ for a fixed $\overline{L}_w = 0.3$, and (b) the dimensionless cutoff wall location $\overline{L}_w$ for a fixed $\overline{h}_w = 0.3$. Each curve is plotted from the coastline to its corresponding interface toe location.

the adoption of the Dupuit–Forchheimer approximation, which assumes predominantly horizontal flow. As demonstrated by the two-dimensional SEAWAT simulations, this assumption may be locally violated in the vicinity of the cutoff wall due to pronounced vertical hydraulic gradients induced by the hydraulic head difference across the wall. However, these vertical flow effects remain localized, and the analytical solution is still able to capture the large-scale freshwater flow behavior and the steady-state interface toe position.

Future research may extend the present analytical framework by relaxing the Dupuit–Forchheimer assumption and by explicitly incorporating transient boundary conditions. Such extensions would allow a more detailed investigation of seawater intrusion processes involving strong vertical flow components and time-dependent forcing, while building upon the physical insights provided by the current steady-state solution.

4. Conclusion

This study developed a novel analytical model for seawater intrusion in coastal confined and unconfined aquifers with a cutoff wall. For the verification of the analytical solutions with an inland constant flux BC, large-scale numerical models based on the variable-density code SEAWAT are employed. Simultaneously, small-scale experimental data extracted from Luyun et al. (2011) is used to validate the analytical solutions under an inland constant head BC. The results of the analytical solution align closely with the results of the numerical solution and experimental data.

The cutoff wall effectiveness in controlling seawater intrusion improves when the coastline and the cutoff wall location are closer, and the cutoff wall opening is narrower. For a fixed cutoff wall location, two critical values of $\overline{h}_w$ emerge. One critical value of $\overline{h}_w = \sqrt{\frac{\kappa(\overline{L}_w-1)+\overline{w}}{1-\kappa}}$ for the confined aquifer and $\overline{h}_w = \sqrt{\frac{(\alpha+1)[\kappa(\overline{L}_w-1)+\overline{w}]}{\alpha(1-\kappa)}}$ for the unconfined aquifer is the lower threshold for controlling seawater intrusion. The other pair is $\overline{h}_w = 1 - \frac{\overline{L}_w+\overline{w}}{2\sqrt{\overline{L}_w(1-\kappa)}}$ and $\overline{h}_w = 1 - \frac{\alpha(\overline{L}_w+\overline{w}+\kappa)+\overline{w}}{2\sqrt{\overline{L}_w\alpha(1-\kappa)}}$ for the confined and unconfined aquifer beyond which the effectiveness of the cutoff wall in containing seawater intrusion is compromised.

The construction of the cutoff wall significantly elevates the hydraulic head upstream of the cutoff wall. The increase in hydraulic head upstream of the cutoff wall becomes more pronounced with the reduction of the cutoff wall opening height and the distance between the cutoff wall location and the coastline.

The models in this study are highly idealized and ignore aquifer heterogeneity, aquifer profiles, and vertical groundwater flow, which are prevalent in the real world. Nevertheless, the findings of this study can provide guidance for the construction of the cutoff wall and enhance the understanding of the mechanism of cutoff walls for controlling seawater intrusion.

CRediT authorship contribution statement

Jiazhi Sun: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Wenlong Shi:** Writing – review & editing, Methodology, Formal analysis. **Huiqiang Wu:** Methodology, Investigation, Formal analysis, Conceptualization. **Jie Yang:** Writing – review & editing, Conceptualization. **Chunhui Lu:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments, Samples, and Data

This work is financially supported by the National Natural Science Foundation of China (U2340212 and 42377084). This work has also been supported by the New Cornerstone Science Foundation through the XPLORER PRIZE.

Appendix A. Derivation of analytical solutions for confined aquifers

In scenario (a), the cutoff wall does not affect the interface zone. Eqs. (1) and (4) give:

$$q_f = K_a h \frac{\partial \varphi^a}{\partial x} \tag{A1}$$

Eq. (2) is written as:

$$h = \alpha\varphi^a - (\alpha+1)H_s + H \tag{A2}$$

Combining Eqs. (A1), (A2), and (3) yields the solution for scenario (a), written as:

$$\varphi^a(x) = \frac{1}{\alpha}\sqrt{\gamma_1 x} + \varphi_s \tag{A3}$$

where

$$\gamma_1 = \frac{2\alpha q_f}{K_a} \tag{A4}$$

and the superscript a denotes the scenario mark. It is worth noting that the shapes of the interface in zone 1 of scenarios (b)–(f) are the same as the shape in scenario (a) as these scenarios share the same governing equation and BC (i.e., Eqs. (A1), (A2), and (3)) in zone 1. That is:

$$\varphi^a(x) = \varphi^b(x) = \varphi^c(x) = \varphi^d(x) = \varphi^e(x) = \varphi^f(x), \quad x \le L_w \tag{A5}$$

The interface toe location is obtained by the combination of Eqs. (6) and (A3):

$$x_t^a = \frac{H^2}{\gamma_1} \tag{A6}$$

Note that Eq. (A6) indicates the interface locations of the aquifer in which the cutoff wall is constructed more inland and the aquifer without a cutoff wall. Thus, the limiting condition of scenario (a) ($L_w \geq x_t^a$) is expressed as:

$$L_w \geq \frac{H^2}{\gamma_1} \tag{A7}$$

In scenario (b), the interface zone is divided into two zones. The expression of the interface in zone 1 is shown in Eq. (A5). In zone 2, Eqs. (1) and (4) give:

$$q_f = (K_a(h - H + h_w) + K_w(H - h_w))\frac{\partial \varphi_2^b}{\partial x} \tag{A8}$$

The continuity equation at the intersection of zones 1 and 2 (i.e., $x = L_w$) is written as:

$$\varphi_1^b(L_w) = \varphi_2^b(L_w) \tag{A9}$$

Combination of Eqs. (A2), (A8), and (A9) gives:

$$\varphi_2^b = \frac{1}{\alpha}\left(\sqrt{\gamma_1(x - L_w) + \lambda_1^2} + \beta_1\right) + \varphi_s \tag{A10}$$

where

$$\beta_1 = (H - h_w)(1 - \kappa), \quad \lambda_1 = \beta_1 - \sqrt{\gamma_1 L_w} \tag{A11}$$

It is worth noting that two solutions can be obtained by Eqs. (2), (A8), and (A9), one of the solutions (i.e., Eq. (A10)) is chosen as the other one is smaller than the hydraulic head of the coastal BC (i.e., φ_s). Substituting Eq. (A10) into Eq. (6) gives the interface toe location:

$$x_t^b = (-1 + 2\kappa)x_t^a + \frac{2}{\gamma_1}\left(\beta_1\sqrt{L_w\gamma_1} + (1 - \kappa)h_w H\right) \tag{A12}$$

The elevation of the interface at the intersection of zone 1 and zone 2 is:

$$\zeta(L_w) = \alpha\varphi_1^a(L_w) - (\alpha + 1)H_s + H \tag{A13}$$

Thus, the limiting conditions of scenario (b), $h_w > \zeta(L_w)$ and $x_t^b < L_w + w$, are written as:

$$H - \sqrt{\gamma_1 L_w} < h_w < \frac{2H(\kappa - 1)\sqrt{\gamma_1 L_w} + \gamma_1(L_w + w) - (2\kappa - 1)H^2}{2(\kappa - 1)\left(\sqrt{\gamma_1 L_w} - H\right)} \tag{A14}$$

Note that Eq. (A14) is applicable only when the value on the left-hand side is smaller than that on the right-hand side. Therefore, the validity of this condition should be checked by comparing the two sides.

In scenario (c), the interface zone is divided into three zones. Expressions of hydraulic head in zones 1 and 2 are the same as expressions in scenario (b), written as:

$$\begin{aligned} &\varphi_1^c(x) = \varphi_1^b(x), \quad x \leq L_w \\ &\varphi_2^c(x) = \varphi_2^b(x), \quad L_w < x \leq L_w + w \end{aligned} \tag{A15}$$

In zone 3, the governing equation and continuity equation, respectively, are:

$$q_f = K_a h\frac{\partial \varphi_3^c}{\partial x} \tag{A16}$$

and

$$\varphi_2^c(L_w + w) = \varphi_3^c(L_w + w) \tag{A17}$$

Combination of Eqs. (A16) and (A17) yields:

$$\varphi_3^c(x) = \frac{1}{\alpha}\sqrt{\gamma_1 x + 2\beta_1\lambda_1 + 2\beta_1\sqrt{\lambda_1^2 + \gamma_1 w}} + \varphi_s \tag{A18}$$

Substituting Eq. (A18) into Eq. (6) gives the interface toe location in scenario (c):

$$x_t^c = x_t^a - 2\beta_1 / \gamma_1\left(\lambda_1 + \sqrt{w\gamma_1 + \lambda_1^2}\right) \tag{A19}$$

Thus, the limiting conditions of scenario (c), $h_w > \zeta(L_w)$ and $x_t^c > L_w + w$, are obtained by Eqs. (A13) and (A19):

$$h_w > \max\left[H - \sqrt{\gamma_1 L_w}, \frac{2H(\kappa - 1)\sqrt{\gamma_1 L_w} + \gamma_1(L_w + w) - (2\kappa - 1)H^2}{2(\kappa - 1)\left(\sqrt{\gamma_1 L_w} - H\right)}\right] \tag{A20}$$

In scenario (d), the interface zone is divided into two zones. Expressions of hydraulic head in zone 1 are referred to as Eq. (A5). Eqs. (1), (4), and (5) give the governing equation and continuity at $x = L_w$, written as:

$$q_f = K_w h\frac{\partial \varphi_2^d}{\partial x} \tag{A21}$$

and

$$\varphi_1^d(L_w) = \varphi_2^d(L_w) \tag{A22}$$

Eqs. (A3), (A5), (A21), and (A22) give the expression of the hydraulic head in zone 2:

$$\varphi_2^d(x) = \frac{1}{\alpha}\sqrt{\frac{\gamma_1 x - (1-\kappa)L_w\gamma_1}{\kappa}} + \varphi_s \tag{A23}$$

As shown in Fig. 2, the interface intersects the cutoff wall bottom (i.e., $z = H$ - h_w) at $x = x_h$. The freshwater flow thickness is written as:

$$h(x_h) = H - h_w = \alpha\varphi_2^d(x_h) - (\alpha+1)H_s + H \tag{A24}$$

Substituting Eq. (A23) into Eq. (A24) yields:

$$x_h = L_w(1-\kappa) + \frac{\kappa(H-h_w)^2}{\gamma_1} \tag{A25}$$

The governing equation of zone 3 and continuity equation at $x = x_h$, respectively, are:

$$q_f = (K_a(h - H + h_w) + K_w(H - h_w))\frac{\partial\varphi_3^d}{\partial x} \tag{A26}$$

and

$$\varphi_2^d(x_h) = \varphi_3^d(x_h) \tag{A27}$$

Eqs. (A3), (A5), (A26), and (A27) give the expression of hydraulic head in zone 3:

$$\varphi_3^d(x) = \frac{\beta_1 + \sqrt{(x-(1-\kappa)L_w)\gamma_1 - \kappa\beta_1(H-h_w)}}{\alpha} + \varphi_s \tag{A28}$$

Substituting Eq. (A28) into Eq. (6) gives the interface toe location in scenario (d):

$$x_t^d = \frac{\kappa H^2 + (1-\kappa)\left(L_w\gamma_1 + {h_w}^2\right)}{\gamma_1} \tag{A29}$$

Thus, the limiting conditions of $h_w < \zeta(L_w)$, $x_h < L_w + w$ and $x_t^d < L_w + w$, based on Eqs. (A13), (A25), and (A29) are expressed as:

$$H - \sqrt{\gamma_1 L_w + \frac{\gamma_1 w}{\kappa}} < h_w < \min\left[H - \sqrt{\gamma_1 L_w}, \sqrt{\left(\frac{L_w + w}{1-\kappa} - L_w\right)\gamma_1 - \frac{\kappa H^2}{1-\kappa}}\right] \tag{A30}$$

In scenario (e), the interface zone is divided into four zones. Expressions of hydraulic head in zones 1–3 are the same as expressions in scenario (d), which are:

$$\begin{aligned} &\varphi_1^e(x) = \varphi^a(x), && x \le L_w \\ &\varphi_2^e(x) = \varphi_2^d(x), && L_w < x \le x_h \\ &\varphi_3^e(x) = \varphi_3^d(x), && x_h < x \le L_w + w \end{aligned} \tag{A31}$$

In zone 4, the governing equation and continuity equation are the same as Eqs. (A16) and (A17), written as:

$$q_f = K_a h\frac{\partial\varphi_4^e}{\partial x} \tag{A32}$$

and

$$\varphi_3^e(L_w + w) = \varphi_4^e(L_w + w) \tag{A33}$$

Eqs. (A3), (A5), (A32), and (A33) give the expression of hydraulic head in zone 4:

$$\varphi_4^e(x) = \frac{1}{\alpha}\sqrt{(x-(1-\kappa)L_w)\gamma_1 + \beta_1\left(\beta_1 - \kappa H_w + 2\sqrt{(\kappa L_w + w)\gamma_1 - \kappa\beta_1 H_w}\right)} + \varphi_s \tag{A34}$$

Substituting Eq. (A34) into Eq. (6) gives the interface toe location in scenario (e):

$$x_t^e = \frac{1}{\gamma_1}\left(H^2(3-2\kappa)\kappa - (1-\kappa)\left(\begin{array}{l} 2H_w\sqrt{(\kappa L_w + w)\gamma_1 - H_w\beta_1\kappa} \\ -h_w(2H - h_w)(1-2\kappa) - L_w\gamma_1 \end{array}\right)\right) \tag{A35}$$

Thus, $h_w < \zeta(L_w)$, $x_h < L_w + w$ and $x_t^e > L_w + w$ can be written as:

$$\max\left[H - \sqrt{\gamma_1 L_w + \frac{\gamma_1 w}{\kappa}}, \sqrt{\left(\frac{L_w + w}{1-\kappa} - L_w\right)\gamma_1 - \frac{\kappa H^2}{1-\kappa}}\right] < h_w < H - \sqrt{\gamma_1 L_w} \tag{A36}$$

In scenario (f), the interface zone is divided into three zones. Expressions of hydraulic head in zones 1 and 2 are the same as expressions in scenario (e), which are:

$$\begin{aligned}\varphi_1^f(x) &= \varphi^a(x), \qquad x \le L_w \\ \varphi_2^f(x) &= \varphi_2^e(x), \quad L_w < x \le L_w + w\end{aligned} \tag{A37}$$

In zone 3, the governing and continuity equations are the same as Eqs. (A32) and (A33). Following a similar procedure with zone 4 in scenario (f), the expression of zone 3 for scenario (f) is obtained:

$$\varphi_3^f(x) = \frac{1}{\alpha}\sqrt{\left(x - \left(1 - \frac{1}{\kappa}\right)w\right)\gamma_1} + \varphi_s \tag{A38}$$

Substituting Eq. (A38) into Eq. (6) gives the interface toe location in scenario (f):

$$x_t^f = x_t^a + w\left(1 - \frac{1}{\kappa}\right) \tag{A39}$$

Thus, the limiting conditions of $h_w < \zeta(L_w)$ and $h_w < \zeta(L_w + w)$ are expressed as:

$$h_w < H - \sqrt{\gamma_1\left(L_w + \frac{w}{\kappa}\right)} \tag{A40}$$

Appendix B. Derivation of Analytical Solutions for Unconfined Aquifers

The derivation of analytical solutions for unconfined aquifers with a cutoff wall is similar to Appendix A.
In scenario (a), Eqs. (1) and (4) give:

$$q_f = K_a h \frac{\partial \varphi^{u,a}}{\partial x} \tag{B1}$$

Eq. (2) is written as:

$$h = (\alpha + 1)(\varphi - H_s) \tag{B2}$$

Combining Eqs. (B1), (B2), and (3) yields the solution for scenario (a):

$$\varphi^{u,a}(x) = \frac{1}{\alpha}\sqrt{\gamma_2 x} + \varphi_s \tag{B3}$$

where

$$\gamma_2 = \frac{2q_f}{(\alpha + 1)K_a} \tag{B4}$$

The interface toe location is obtained by solving Eq. (6):

$$x_t^{u,a} = \frac{{H_s}^2}{\gamma_2 \alpha^2} \tag{B5}$$

Thus, the limiting condition of scenario (a) is:

$$L_w \ge \frac{{H_s}^2}{\gamma_2 \alpha^2} \tag{B6}$$

The hydraulic head in zone 1 of all scenarios is identical for unconfined aquifers. That is:

$$\varphi^{u,a}(x) = \varphi^{u,b}(x) = \varphi^{u,c}(x) = \varphi^{u,d}(x) = \varphi^{u,e}(x) = \varphi^{u,f}(x), \quad x \le L_w \tag{B7}$$

In scenario (b), the hydraulic head in zone 1 is shown in Eq. (B7). In zone 2, Eqs. (1) and (4) become:

$$q_f = \left(K_a\left(h - \varphi_2^{u,b} + h_w\right) + K_w\left(\varphi_2^{u,b} - h_w\right)\right)\frac{\partial \varphi_2^{u,b}}{\partial x} \tag{B8}$$

Continuity equation at the intersection (i.e., $x = L_w$) is written as:

$$\varphi_1^{u,b}(L_w) = \varphi_2^{u,b}(L_w) \tag{B9}$$

The combination of Eqs. (2), (B8), and (B9) gives:

$$\varphi_2^{u,b} = \sqrt{\frac{\gamma_2(\alpha + 1)(x - L_w)}{\alpha + \kappa} + \left(\beta_2 - \sqrt{L_w \gamma_2}\right)^2} + \beta_2 + \varphi_s \tag{B10}$$

where

$$\beta_2 = \frac{(H_s - h_w)(1 - \kappa)}{\alpha + \kappa}, \quad \lambda_2 = \beta_2 - \sqrt{L_w \gamma_2} \tag{B11}$$

The interface toe location is obtained by substituting Eq. (B10) into Eq. (6):

$$x_t^{u,b} = \kappa x_t^{u,a} + \frac{\kappa - 1}{(1+\alpha)\gamma_2}\left(\frac{1}{\alpha}\left({H_s}^2 - {h_w}^2\right) + \sqrt{L_w\gamma_2}\left(2(h_w - H_s) - \sqrt{L_w\gamma_2}\right)\right) \tag{B12}$$

The elevation of the interface at the intersection of zones 1 and 2 is:

$$\zeta(L_w) = (\alpha+1)H_s - \alpha\varphi_1^{u,a}(L_w) \tag{B13}$$

Thus, the limiting conditions of $h_w > \zeta(L_w)$ and $x_t^{u,b} < L_w + w$ are written as:

$$\max\left[H_s - \alpha\sqrt{L_w\gamma_2}, \frac{H_s}{2} - \frac{\alpha(\alpha+\kappa)\sqrt{\gamma_2 L_w} + H_s\kappa(1+\alpha)}{2(1-\kappa)\alpha}\right] < h_w < H_s - \frac{(\kappa+\alpha)\left(H_s + \alpha\sqrt{\gamma_2 L_w}\right)}{2(1-\kappa)\alpha} + \frac{w\gamma_2\alpha(1+\alpha)}{2(1-\kappa)\left(H_s - \alpha\sqrt{\gamma_2 L_w}\right)} \tag{B14}$$

Note that Eq. (B14) is active when the value on the left-hand side is smaller than the value on the right-hand side.

In scenario (c), the hydraulic heads in zones 1 and 2 are written as:

$$\begin{aligned}\varphi_1^{u,c}(x) &= \varphi_1^{u,b}(x), \quad x \le L_w \\ \varphi_2^{u,c}(x) &= \varphi_2^{u,b}(x), \quad L_w < x \le L_w + w\end{aligned} \tag{B15}$$

In zone 3, the governing equation and continuity equation are:

$$q_f = K_a h\frac{\partial\varphi_3^{u,c}}{\partial x} \tag{B16}$$

and

$$\varphi_2^{u,c}(L_w + w) = \varphi_3^{u,c}(L_w + w) \tag{B17}$$

Eqs. (2), (B16), and (B17) give:

$$\varphi_3^{u,c}(x) = \sqrt{\gamma_2\left(x + \frac{(1-\kappa)w}{\kappa+\alpha}\right) + 2\beta_2\left(\lambda_2 + \sqrt{{\lambda_2}^2 + \frac{(1+\alpha)w\gamma_2}{\kappa+\alpha}}\right)} + \varphi_s \tag{B18}$$

The interface toe location is obtained by substituting Eq. (B18) into Eq. (6):

$$x_t^{u,c} = x_t^{u,a} - \frac{(1-\kappa)w}{\kappa+\alpha} - 2\beta_2\left(\frac{\lambda_2}{\gamma_2} + \sqrt{\frac{(1+\alpha)w}{(\kappa+\alpha)\gamma_2} + \left(\frac{\lambda_2}{\gamma_2}\right)^2}\right) \tag{B19}$$

Thus, $h_w > \zeta(L_w)$ and $x_t^{u,c} > L_w + w$ gives:

$$h_w > \max\left[H_s - \alpha\sqrt{\gamma_2 L_w}, H_s - \frac{(\kappa+\alpha)\left(H_s + \alpha\sqrt{\gamma_2 L_w}\right)}{2(1-\kappa)\alpha} + \frac{w\gamma_2\alpha(1+\alpha)}{2(1-\kappa)\left(H_s - \alpha\sqrt{\gamma_2 L_w}\right)}\right] \tag{B20}$$

In scenario (d), the hydraulic head in zone 1 is shown in Eq. (B7). In zone 2, the governing equation and continuity equation are:

$$q_f = K_w h\frac{\partial\varphi_2^{u,d}}{\partial x} \tag{B21}$$

and

$$\varphi_1^{u,d}(L_w) = \varphi_2^{u,d}(L_w) \tag{B22}$$

The combination of Eqs. (2), (B21), and (B22) gives:

$$\varphi_2^{u,d}(x) = \sqrt{\frac{(x - (1-\kappa)L_w)\gamma_2}{\kappa}} + \varphi_s \tag{B23}$$

The interface intersects the cutoff wall bottom at $x = x_h$ in zone 2. The interface elevation is expressed as:

$$\zeta(x_h) = h_w = -\alpha\varphi_2^d(x_h) + (\alpha+1)H_s \tag{B24}$$

Substituting Eq. (B23) into Eq. (B24) gives:

$$x_h = L_w(1-\kappa) + \frac{\kappa(H_s - h_w)^2}{\alpha^2\gamma_2} \tag{B25}$$

In zone 3, the governing equation and continuity equation are:

$$q_f = \left(K_a\left(h_w - \varphi_3^{u,d} + h\right) + K_w\left(\varphi_3^{u,d} - h_w\right)\right)\frac{\partial\varphi_3^{u,d}}{\partial x} \tag{B26}$$

and

$$\varphi_2^{u,d}(x_h) = \varphi_3^{u,d}(x_h) \tag{B27}$$

Eqs. (B26) and (B27) combining with Eq. (2) give:

$$\varphi_3^{u,d}(x) = \sqrt{\frac{\gamma_2(1+\alpha)}{\kappa+\alpha}(x-(1-\kappa)L_w) - \frac{\kappa(1+\alpha)}{\alpha(1-\kappa)}\beta_2^2} + \beta_2 + \varphi_s \tag{B28}$$

The interface toe location is obtained by substituting Eq. (B28) into Eq. (6):

$$x_t^{u,d} = \kappa x_t^{u,a} + (1-\kappa)L_w + \frac{(1-\kappa){h_w}^2}{\alpha(1+\alpha)\gamma_2} \tag{B29}$$

Thus, $h_w < \zeta(L_w)$, $x_h < L_w + w$ and $x_t^{u,d} < L_w + w$ gives:

$$H_s - \alpha\sqrt{\gamma_2\left(\frac{w}{\kappa}+L_w\right)} < h_w < \min\left[H_s - \alpha\sqrt{L_w\gamma_2}, \sqrt{\frac{(\alpha+1)\left({H_s}^2\kappa - \alpha^2\gamma_2(\kappa L_w + w)\right)}{\alpha(\kappa-1)}}\right] \tag{B30}$$

In scenario (e), the hydraulic head of zones 1–3 is expressed as:

$$\begin{aligned} &\varphi_1^{u,e}(x) = \varphi_1^{u,d}(x) && x \le L_w \\ &\varphi_2^{u,e}(x) = \varphi_2^{u,d}(x) && L_w < x \le x_h \\ &\varphi_3^{u,e}(x) = \varphi_3^{u,d}(x) && x_h < x \le L_w + w \end{aligned} \tag{B31}$$

In zone 4, the governing equation and continuity equation are:

$$q_f = K_a h \frac{\partial \varphi_4^{u,e}}{\partial x} \tag{B32}$$

and

$$\varphi_3^{u,e}(L_w + w) = \varphi_4^{u,e}(L_w + w) \tag{B33}$$

Eqs. (B32) and (B33) combining with Eq. (2) give:

$$\varphi_4^{u,e}(x) = \sqrt{\gamma_2 x + \frac{\kappa-1}{\alpha(\kappa+\alpha)^2}\left(\frac{(H_s-h_w)^2(\kappa-\alpha+2\kappa\alpha) + \alpha(\kappa+\alpha)(L_w\alpha - w)\gamma_2 - 2(H_s-h_w)}{\sqrt{\alpha(1+\alpha)\left(\alpha\gamma_2(\kappa L_w + w)(\kappa+\alpha) - (H_s-h_w)^2(1-\kappa)\kappa\right)}}\right)} + \varphi_s \tag{B34}$$

The interface toe location is obtained by substituting Eq. (B34) into Eq. (6):

$$x_t^{u,e} = \frac{\kappa {H_s}^2(1+\alpha)(3\alpha+\kappa(1-2\alpha))}{\alpha^2(\kappa+\alpha)^2\gamma_2} + \frac{(1-\kappa)\alpha}{\alpha^2(\kappa+\alpha)^2\gamma_2}\left(\frac{({h_w}^2 - 2H_s h_w)(\kappa-\alpha+2\kappa\alpha) + \alpha\gamma_2(\kappa+\alpha)(\alpha L_w - w) -}{2H_w\sqrt{\alpha(1+\alpha)\left(\alpha\gamma_2(\kappa+\alpha)(\kappa L_w + w) - \kappa {H_w}^2(1-\kappa)\right)}}\right) \tag{B35}$$

Thus, $h_w < \zeta(L_w)$, $x_h < L_w + w$ and $x_t^{u,e} > L_w + w$ gives:

$$\max\left[H_s - \alpha\sqrt{\left(\frac{w}{\kappa}+L_w\right)\gamma_2}, \sqrt{\frac{(\alpha+1)\left({H_s}^2\kappa - \alpha^2\gamma_2(\kappa L_w + w)\right)}{\alpha(\kappa-1)}}\right] < h_w < H_s - \alpha\sqrt{L_w\gamma_2} \tag{B36}$$

In scenario (f), the hydraulic heads in zones 1 and 2 are the same as those in scenario (e), written as:

$$\begin{aligned} &\varphi_1^{u,f}(x) = \varphi^{u,a}(x), && x \le L_w \\ &\varphi_2^{u,f}(x) = \varphi_2^{u,e}(x), && L_w < x \le L_w + w \end{aligned} \tag{B37}$$

The expression of the hydraulic head in zone 3 is obtained by solving Eqs. (B32) and (B33), which are identical in scenarios (e) and (f):

$$\varphi_3^{u,f}(x) = \frac{1}{\alpha}\sqrt{\left(x - \left(1-\frac{1}{\kappa}\right)w\right)\gamma_2} + \varphi_s \tag{B38}$$

The interface toe location is obtained by substituting Eq. (B38) into Eq. (6):

$$x_t^{u,f} = x_t^{u,a} + w\left(1-\frac{1}{\kappa}\right) \tag{B39}$$

Thus, the limiting conditions of $h_w < \zeta(L_w)$, and $h_w < \zeta(L_w + w)$ are written as:

$$h_w < H_s - \alpha\sqrt{\gamma_2\left(L_w + \frac{w}{\kappa}\right)} \tag{B40}$$

Appendix C. Transformation of an Inland Constant-Flux BC to a Constant-Head BC

For the groundwater flow between the interface toe and the inland boundary, we have:

$$q_f = K_a h \frac{\partial \varphi}{\partial x} \tag{C1}$$

The thickness of the flow path equals the hydraulic head in an unconfined aquifer, which is written as:

$$h = \varphi \tag{C2}$$

The inland BC is specified as a constant head BC:

$$x = L, \quad \varphi = H_f \tag{C3}$$

Combining with Eqs. (C2) and (C3), the solution of Eq. (C1) is expressed as:

$$\varphi(x) = \sqrt{H_f^2 + \frac{2q_f}{K_a}(x - L)} \tag{C4}$$

At the interface toe, the hydraulic head is:

$$x = x_t, \quad \varphi = \frac{\alpha + 1}{\alpha} H_s \tag{C5}$$

Substituting Eq. (C5) into (C4) gives:

$$q_f = \frac{K_a}{2(L - x_t)} \left(H_f^2 - H_s^2 \frac{(\alpha + 1)^2}{\alpha^2} \right) \tag{C6}$$

Eq. (C6) describes the relationship between the inland constant head and the inland constant flux. Consequently, the interface toe location in an unconfined aquifer with an inland constant head BC can be determined by substituting Eq. (C6) into the solutions for inland constant flux BC.

Appendix D. Validation under zero-dispersion conditions

To further evaluate the validity of the proposed analytical solution independently of dispersion effects, additional numerical simulations are conducted under zero-dispersion conditions. The comparison between the numerical and analytical solutions is shown in Fig. D1. In these simulations, longitudinal and transverse dispersivity as well as molecular diffusion are set to zero, such that solute transport is governed solely by advection.

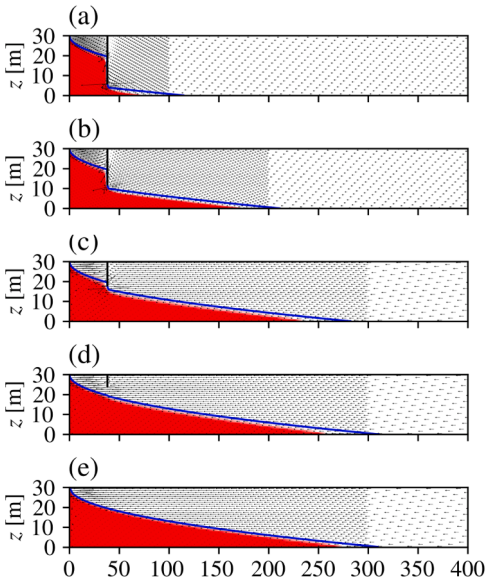


Fig. D1. Comparison between analytical and numerical solutions under zero-dispersion conditions ($L_w = 38$ m). Panels (a)–(e) correspond to $h_w = 6$, 12, 18, 24, and 30 m, respectively.

To minimize numerical dispersion, local grid refinement (i.e., a grid length of 0.25 m) is applied in the vicinity of the cutoff wall and the interface zone. The refined domain extends from 0 to 100 m in panel (a), to 200 m in panel (b), and to 300 m in panels (c)–(e), corresponding to the increasing inland extent of seawater intrusion. The wall thickness is resolved by four grids to ensure an adequate representation of the near-wall flow field. Outside these regions, coarser discretization is retained.

Minor deviations remain between the analytical and numerical interfaces even under zero-dispersion conditions. These discrepancies are primarily attributed to the inherent limitation of the Dupuit–Forchheimer approximation in capturing local two-dimensional flow effects in the interface and toe regions.

Data availability

No data was used for the research described in the article.

References

Abdoulhalik, A., Ahmed, A.A., 2017. The effectiveness of cutoff walls to control saltwater intrusion in multi-layered coastal aquifers: experimental and numerical study. J. Environ. Manage. 199, 62–73. https://doi.org/10.1016/j.jenvman.2017.05.040.

Anwar, H.O., 1983. The effect of a subsurface barrier on the conservation of freshwater in coastal aquifers. Water Res. 17, 1257–1265. https://doi.org/10.1016/0043-1354(83)90250-6.

Badon Ghyben, W., 1888. Nota in Verband Met De Voorgenomen Putboring Nabij Amsterdam. Tijdschr. Kon. Inst. Ing., Gravenhage, Netherlands.

Bakker, M., Post, V., Langevin, C.D., Hughes, J.D., White, J.T., Starn, J.J., Fienen, M.N., 2016. Scripting MODFLOW model development using Python and FloPy. Ground Water 54, 733–739. https://doi.org/10.1111/gwat.12413.

Brovelli, A., Mao, X., Barry, D.A., 2007. Numerical modeling of tidal influence on density-dependent contaminant transport. Water Resour. Res. 43. https://doi.org/10.1029/2006WR005173.

Chang, Q., Zheng, T., Chen, Y., Zheng, X., Walther, M., 2021. Influence of inland freshwater influx on the natural desalination of coastal aquifers with a cutoff wall. Desalination 499, 114863. https://doi.org/10.1016/j.desal.2020.114863.

Dror, I., Berkowitz, B., Gorelick, S.M., 2004. Effects of air injection on flow through porous media: observations and analyses of laboratory-scale processes. Water Resour. Res. 40. https://doi.org/10.1029/2003WR002960.

Ebeling, P., Händel, F., Walther, M., 2019. Potential of mixed hydraulic barriers to remediate seawater intrusion. Sci. Total Environ. 693, 133478. https://doi.org/10.1016/j.scitotenv.2019.07.284.

Foster, S., Garduno, H., Evans, R., Olson, D., Tian, Y., Zhang, W., Han, Z., 2004. Quaternary Aquifer of the North China Plain-assessing and achieving groundwater resource sustainability. Hydrogeol. J. 12, 81–93. https://doi.org/10.1007/s10040-003-0300-6.

Herrera-García, G., Ezquerro, P., Tomás, R., Béjar-Pizarro, M., López-Vinielles, J., Rossi, M., Mateos, R.M., Carreón-Freyre, D., Lambert, J., Teatini, P., Cabral-Cano, E., Erkens, G., Galloway, D., Hung, W.C., Kakar, N., Sneed, M., Tosi, L., Wang, H., Ye, S., 2021. Mapping the global threat of land subsidence. Science 371, 34–36. https://doi.org/10.1126/science.abb8549.

Herzberg, A., 1901. Die wasserversorgung einiger nordseebader. J. Gasbeleucht. Wasserversorg. 44 (815–819), 842–844.

Ishida, S., Kotoku, M., Abe, E., Fazal, M.A., Tsuchihara, T., Imaizumi, M., 2003. Construction of subsurface dams and these impact on the environment. Mater. Geoenviron. 50, 149–152.

Jamali, I.A., Olofsson, B., Mörtberg, U., 2013. Locating suitable sites for the construction of subsurface dams using GIS. Environ. Earth Sci. 70, 2511–2525. https://doi.org/10.1007/s12665-013-2295-1.

Japan Green Resources Agency, 2004. Technical Reference for Effective Groundwater Development. Japan Green Resources Agency, Saiwai-ku, Kawasaki, Kanagawa Prefecture, Japan.

Kacimov, A.R., Sherif, M.M., Perret, J.S., Al-Mushikhi, A., 2009. Control of sea-water intrusion by salt-water pumping: coast of Oman. Hydrogeol. J. 17, 541–558. https://doi.org/10.1007/s10040-008-0425-8.

Kaleris, V.K., Ziogas, A.I., 2013. The effect of cutoff walls on saltwater intrusion and groundwater extraction in coastal aquifers. J. Hydrol. 476, 370–383. https://doi.org/10.1016/j.jhydrol.2012.11.007.

Ketabchi, H., Mahmoodzadeh, D., Ataie-Ashtiani, B., Simmons, C.T., 2016. Sea-level rise impacts on seawater intrusion in coastal aquifers: review and integration. J. Hydrol. 535, 235–255. https://doi.org/10.1016/j.jhydrol.2016.01.083.

Land, M., Reichard, E.G., Crawford, S.M., Everett, R.R., Newhouse, M.W., Williams, C.F., 2004. Ground-water quality of coastal aquifer systems in the west coast basin, Los Angeles County, California, 1999–2002. US Geol. Surv. Sci. Investig. Rep. 5067, 88.

Langevin, C.D., Guo, W., 2006. MODFLOW/MT3DMS-based simulation of variable-density ground water flow and transport. Ground Water 44, 339–351. https://doi.org/10.1111/j.1745-6584.2005.00156.x.

Liu, J., Zheng, C., Zheng, L., Lei, Y., 2008. Ground water sustainability: methodology and application to the North China Plain. Groundwater 46, 897–909. https://doi.org/10.1111/j.1745-6584.2008.00486.x.

Lu, C., Shi, W., Xin, P., Wu, J., Werner, A.D., 2017. Replenishing an unconfined coastal aquifer to control seawater intrusion: injection or infiltration? Water Resour. Res. 53, 4775–4786. https://doi.org/10.1002/2016wr019625.

Lu, C., Werner, A.D., 2013. Timescales of seawater intrusion and retreat. Adv. Water Resour. 59, 39–51. https://doi.org/10.1016/j.advwatres.2013.05.005.

Lu, C., Werner, A.D., Simmons, C.T., Robinson, N.I., Luo, J., 2013. Maximizing net extraction using an injection-extraction well pair in a coastal aquifer. Ground Water 51, 219–228. https://doi.org/10.1111/j.1745-6584.2012.00973.x.

Lu, C., Xin, P., Kong, J., Li, L., Luo, J., 2016. Analytical solutions of seawater intrusion in sloping confined and unconfined coastal aquifers. Water Resour. Res. 52, 6989–7004. https://doi.org/10.1002/2016wr019101.

Lu, C., Xin, P., Li, L., Luo, J., 2015. Seawater intrusion in response to sea-level rise in a coastal aquifer with a general-head inland boundary. J. Hydrol. 522, 135–140. https://doi.org/10.1016/j.jhydrol.2014.12.053.

Luyun, R., Momii, K., Nakagawa, K., 2011. Effects of recharge wells and flow barriers on seawater intrusion. Ground Water 49, 239–249. https://doi.org/10.1111/j.1745-6584.2010.00719.x.

Luyun, R., Momii, K., Nakagawa, K., 2009. Laboratory-scale saltwater behavior due to subsurface cutoff wall. J. Hydrol. 377, 227–236. https://doi.org/10.1016/j.jhydrol.2009.08.019.

Mahesha, A., 2001. Effect of strip recharge on sea water intrusion into aquifers. Hydrol. Sci. J. 46, 199–210. https://doi.org/10.1080/02626660109492816.

Mahesha, A., 1996. Transient effect of battery of injection wells on seawater intrusion. J. Hydraul. Eng. 122, 266–271. https://doi.org/10.1061/(ASCE)0733-9429(1996)122:5(266).

Michael, H.A., Post, V.E.A., Wilson, A.M., Werner, A.D., 2017. Science, society, and the coastal groundwater squeeze. Water Resour. Res. 53, 2610–2617. https://doi.org/10.1002/2017WR020851.

Pool, M., Carrera, J., 2011. A correction factor to account for mixing in Ghyben-Herzberg and critical pumping rate approximations of seawater intrusion in coastal aquifers. Water Resour. Res. 47. https://doi.org/10.1029/2010wr010256.

Post, V.E.A., Groen, J., Kooi, H., Person, M., Ge, S., Edmunds, W.M., 2013. Offshore fresh groundwater reserves as a global phenomenon. Nature 504, 71–78. https://doi.org/10.1038/nature12858.

Rohatgi, A., 2021. Webplotdigitizer: version 4.5.

Sheahan, N.T., 1977. Injection/extraction well system–a unique seawater intrusion barrier. Ground Water 15, 32–50. https://doi.org/10.1111/j.1745-6584.1977.tb03150.x.

Sherif, M.M., Hamza, K.I., 2001. Mitigation of seawater intrusion by pumping brackish water. Transp. Porous. Media 43, 29–44. https://doi.org/10.1023/A:1010601208708.

Shi, L., Lu, C., Ye, Y., Xie, Y., Wu, J., 2020. Evaluation of the performance of multiple-well hydraulic barriers on enhancing groundwater extraction in a coastal aquifer. Adv. Water Resour. 144, 103704. https://doi.org/10.1016/j.advwatres.2020.103704.

Strack, O.D.L., 1976. A single-potential solution for regional interface problems in coastal aquifers. Water Resour. Res. 12, 1165–1174. https://doi.org/10.1029/wr012i006p01165.

Strack, O.D.L., Ausk, B.K., 2015. A formulation for vertically integrated groundwater flow in a stratified coastal aquifer. Water Resour. Res. 51, 6756–6775. https://doi.org/10.1002/2015wr016887.

Sun, D., Semprich, S., 2013. Using compressed air injection to control seawater intrusion in a confined coastal aquifer. Transp. Porous. Media 100, 259–278. https://doi.org/10.1007/s11242-013-0215-1.

Voss, C.I., Souza, W.R., 1987. Variable density flow and solute transport simulation of regional aquifers containing a narrow freshwater-saltwater transition zone. Water Resour. Res. 23, 1851–1866. https://doi.org/10.1029/WR023i010p01851.

Werner, A.D., Bakker, M., Post, V.E.A., Vandenbohede, A., Lu, C., Ataie-Ashtiani, B., Simmons, C.T., Barry, D.A., 2013. Seawater intrusion processes, investigation and management: recent advances and future challenges. Adv. Water Resour. 51, 3–26. https://doi.org/10.1016/j.advwatres.2012.03.004.

Wu, H., Lu, C., Kong, J., Werner, A.D., 2020. Preventing seawater intrusion and enhancing safe extraction using finite-length, impermeable subsurface barriers: 3D analysis. Water Resour. Res. 56, e2020WR027792. https://doi.org/10.1029/2020WR027792.

Xie, Y., Simmons, C.T., Werner, A.D., Ward, J.D., 2010. Effect of transient solute loading on free convection in porous media. Water Resour. Res. 46. https://doi.org/10.1029/2010WR009314.

Yang, X., Cao, H., Werner, A.D., Luo, J., Lu, C., 2024. Experimental and numerical investigation of the effect of air injection on seawater intrusion mitigation. J. Hydrol., 131850 https://doi.org/10.1016/j.jhydrol.2024.131850.

Zheng, T., Zheng, X., Sun, Q., Wang, L., Walther, M., 2020. Insights of variable permeability full-section wall for enhanced control of seawater intrusion and nitrate contamination in unconfined aquifers. J. Hydrol. 586, 124831.